

The Unequal Weight of Genes: How Early Life Environments Modify the Effects of Genetic Predispositions

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Abstract

Children from poorer families show a stronger link between genetic predictors and body weight. The dominant interpretation is that disadvantaged environments amplify children’s own genetic risk for obesity. Previous research has relied on standard polygenic indices, which conflate the child’s directly inherited genetic effects with the effects of parents’ genotypes operating through the household environment, and thus has been unable to disentangle the two channels. I use linked parent–child genotypes from 5,446 children in the UK Millennium Cohort Study to compare a standard polygenic index for body mass index (from a 700,000-individual population GWAS) with a direct-effects polygenic index (from an 82,000-family GWAS that isolates inherited variation). The standard index reproduces the documented socioeconomic gradient; the direct-effects index’s gradient is statistically indistinguishable from zero. The gradient is thus carried by the standard index and not by the direct-effects index. The paper makes three contributions. First, it introduces the Standard-versus-DGE design for separating direct-effect moderation (H1) from family-channel moderation (H2) within a single sample. Second, it establishes that the widely cited interpretation of the BMI gradient—that disadvantage amplifies children’s own genetic risk—is not supported by the evidence: the gradient is equally consistent with a family-channel reading, and a formal restriction test cannot distinguish the two. Third, it identifies precisely why the design cannot yet decide between them, and turns that into a result the field needs as family-GWAS indices come into use: at currently available discovery sizes a family-GWAS index predicts the outcome too weakly to separate the channels, so a null direct-effects interaction is not evidence that environments leave children’s own genetic risk untouched. Applying the same comparison to smoking yields a null for both indices, confirming that the BMI gradient is a real, trait-specific phenomenon and not an artefact of the design. Whether disadvantage amplifies children’s own genetic risk—and hence whether genetic screening of children has any rationale—is therefore an open question with a clear data requirement: a substantially larger family GWAS.

1 Introduction

Socioeconomic gradients in childhood obesity are large, persistent, and a major public-health concern in most high-income countries ([Autret and Bekelman, 2024](#)), with downstream consequences for health, labour market outcomes, and wellbeing ([Lindeboom et al., 2010](#); [Segal et al., 2021](#)). An influential body of work using polygenic indices (PGIs) has consistently shown that the PGI–BMI association is substantially stronger among children from lower-SES than higher-SES households

(Hüls et al., 2021; Bann et al., 2022). A PGI is a composite genetic score that aggregates many common variants into a single BMI predictor. The PGI-SES interaction is typically read as evidence that disadvantaged environments “unlock” or “amplify” children’s genetic susceptibility to weight gain. This reading carries an implicit policy prescription: compress the environmental gradient and the genetic gradient will compress with it.

This interpretation glosses over an identification problem that the conventional design cannot solve. A standard PGI is built from population-GWAS weights that bundle three distinct genetic channels at each variant (Kong et al., 2018): the child’s own direct effect on BMI, the indirect effects of parental genotypes operating through parenting and household environments (Wertz et al., 2019; Breinholt and Conley, 2023), and residual population stratification. A steeper PGI-BMI slope at low SES is therefore consistent with at least two causal stories that have different implications. Under direct-effect moderation (H1), disadvantage amplifies the within-child translation of genetic risk into body weight: the same alleles produce larger BMI differences at low SES. Under family-channel moderation (H2), the interaction is an artefact of what a standard PGI measures. A child’s PGI partly reflects inherited parental genotypes, so it correlates with parental behaviours, household food environments, and resource allocation. If those family channels are stronger at low SES, a standard PGI will appear to predict BMI more steeply among disadvantaged children even when children’s own biology responds identically to the environment.

The two stories are easy to confuse precisely because BMI is the trait for which they look most alike at first glance. Within-sibship and family-GWAS analyses have shown that for socially-mediated traits like educational attainment, the population-GWAS coefficient is substantially inflated by family-level genetic channels relative to the within-family estimate (Howe et al., 2022; Tan et al., 2026), which makes the standard PGI an obviously contaminated measure of the child’s own effect. For BMI, in contrast, this average gap is statistically indistinguishable from zero (Howe et al., 2022; Tan et al., 2026), and the literature has read this main-effect agreement as evidence that family channels are quantitatively unimportant for BMI. The $\text{PGI} \times \text{SES}$ interaction is the place where that reading breaks down: an indirect channel small on average can still be socioeconomically patterned, because household food environments, parental health behaviours, and resources vary precisely along the SES dimension. BMI is therefore the trait where the bundle matters not because the bundle is large but because it is patterned by the moderator under study.

This paper separates the two channels empirically. Using linked child-mother-father genotypes in the UK Millennium Cohort Study, I construct two polygenic indices for BMI in the same children and compare how each moderates the BMI-SES association. The Standard PGI uses the population-GWAS weights of Yengo et al. (2018b) ($N \approx 700,000$), which aggregate all three channels. The direct genetic effects (DGE) PGI uses the family-GWAS weights of Tan et al. (2026), whose SNP-level estimates isolate the child’s own causal alleles by exploiting within-family Mendelian segregation at the discovery stage. The contrast between the SES interactions of the two predictors is designed to separate which channel the observed $\text{PGI} \times \text{SES}$ gradient loads on; maternal and paternal PGIs enter as controls to absorb residual family-level genetic background. To my knowledge, this is the first application of a Standard-versus-DGE contrast to the $\text{PGI} \times \text{SES}$ gradient for childhood BMI.

The analysis runs in three steps. First, the canonical Standard $\text{PGI} \times \text{SES}$ interaction reproduces the well-documented negative gradient: the implied PGI-BMI slope is 0.28 BMI units per standard deviation steeper in the lowest income quintile than in the highest. Because the Standard PGI bundles three channels, the gradient is real but its causal attribution is ambiguous. Second, re-

placing the Standard PGI with the DGE PGI—whose weights isolate direct effects at the discovery stage—collapses the interaction to a coefficient indistinguishable from zero in every specification. Third, putting both PGIs into a single joint regression and testing the restriction that pure H1 implies—that the two interactions stand in the ratio $r = \sqrt{R_{\text{Std}}^2/R_{\text{DGE}}^2}$ —cannot reject pure H1 at any income quintile. The obstacle is precision: the DGE PGI explains only about 0.04% of the variance of BMI, so its SES interaction is too imprecisely estimated for the contrast to separate the channels. This is itself a result, and one the field needs as family-GWAS indices come into use—it shows that a null family-GWAS interaction cannot be read as the absence of direct-effect moderation. What the contrast does establish is clear: the gradient is present with the Standard PGI and absent with the DGE PGI. Three further probes pin down that finding: age heterogeneity locates the Standard gradient in late adolescence; the quintile profile is monotone for the Standard PGI and flat for the DGE PGI; and the same Standard-versus-DGE comparison applied to smoking yields a clean null for both predictors, establishing that the BMI pattern is a real, trait-specific phenomenon and not a mechanical artefact of regressing any PGI on any socially patterned outcome. The cross-sectional SDQ comparison shows a qualitatively different pattern from BMI, confirming that the design captures something trait-specific.

Two conclusions follow. First, the widely cited reading of the BMI gradient—that disadvantage amplifies children’s own genetic risk—is not established by the evidence: the gradient is carried by the Standard PGI and not the DGE PGI, and is equally consistent with a family-channel reading. Second, the question of which channel operates is now precisely posed, with a precisely known requirement for settling it: a family GWAS large enough to give the DGE PGI the precision the contrast needs. The smoking contrast, null for both predictors, and the qualitatively different SDQ contrast confirm that the BMI pattern is trait-specific.

A separate robustness exercise addresses concerns about Mendelian imputation of missing parental genotypes. Restricting the sample to families with both parents directly genotyped—removing any contribution from imputed parental data—preserves the main BMI result and in several specifications strengthens it: the Standard PGI $\times$ SES interaction roughly doubles in the trio sample while the DGE interaction remains essentially zero, so imputation works in the conservative direction (toward zero) and cannot explain the Standard-vs-DGE contrast.

These findings have two implications. First, the case for screening children by genetic risk—the policy prescription the literature’s reading carries—requires that reading to hold, and this paper shows it has not been established: the gradient is equally consistent with operating through parental genotypes and the household, so it does not, on its own, license genetic stratification of children. Second, and as a general methodological point, the Standard-versus-DGE contrast is the right design for separating biological gene-by-environment modification from family-channel artefacts in PGI $\times$ E interactions—and this paper establishes the discovery-size threshold the design requires, showing that a null family-GWAS interaction at current sizes is not evidence that no such modification exists.

The remainder of the paper is organised as follows. §2 reviews how prior observational and quasi-experimental designs have approached the PGI $\times$ SES question and why none implements the contrast this paper makes. §3 develops the mathematical framework end-to-end: the genetic foundations from meiosis to the genotype random variable, the population GWAS and its three-channel bias decomposition, the family-based GWAS that identifies direct genetic effects, the gene-environment estimating equation, and the hypotheses H1 and H2 with their observable predictions. §§4–5

describe the data and the regression specifications. §6 reports the results, including the H1/H2 contrast, the test of pure H1 that bounds what the contrast can identify, and the cross-outcome comparison. §7 discusses the findings and concludes.

2 The identification problem in the existing literature

The negative PGI $\times$ SES gradient for childhood BMI is robustly documented. In European-ancestry observational cohorts, the slope of BMI on a standard PGI is consistently steeper at lower socioeconomic status, replicating across countries, age ranges, and PGI specifications (Hüls et al., 2021; Bann et al., 2022). Differences across studies in the precise magnitude track differences in the age range, cohort context, and phenotype construction (continuous BMI versus overweight classes or trajectories; de Roo et al., 2023), but the qualitative pattern is stable. The H1/H2 question is confounded in any analysis that uses a standard population-GWAS PGI because the standard PGI bundles the direct-effect, indirect-effect, and stratification channels (Kong et al., 2018; Biroli et al., 2026). A non-zero $\hat{\beta}_{GS}$ in a population-PGI regression is therefore consistent with H1, with H2, with residual stratification (Mason et al., 2020), or with any mixture (Carter et al., 2022; Pereira et al., 2022). Several recent papers identify this ambiguity explicitly: van den Berg et al. (2025) note that their gene–environment estimates “may be capturing the additional exposure [to confectionery] driven by the parental genetic ‘predisposition’ [for sugar]”; Uchikoshi and Conley (2021) call for designs that distinguish direct genetic mechanisms from genetic nurture in PGI interaction estimates; Pereira et al. (2022) characterise family-based gene–environment analysis as the literature’s “ideal experiment” that has yet to be implemented for health outcomes.

Quasi-experimental designs answer a related but different question and do not close the gap. Studies that exploit compulsory schooling reforms (Barcellos et al., 2018), drinking-age discontinuities (Fletcher and Lu, 2021), prenatal exposures (van den Berg et al., 2025; Dias Pereira et al., 2025), or within-family investment variation (Muslimova et al., 2024) estimate treatment-effect heterogeneity by genotype: how the causal effect of an external shock varies across the polygenic distribution. This is conceptually distinct from asking whether SES itself modifies the expression of children’s own genetic risk. The reported magnitudes are heterogeneous across traits and designs (Baker et al., 2024; Furuya et al., 2022), and the distinction between “treatment heterogeneity by genotype” and “SES moderation of direct genetic effects” is often conflated when the results are summarised.

Within-family designs in twins and siblings reduce shared-family confounding but do not solve the channel-attribution problem when SES is used as the environment (Amin et al., 2018). The few PGI $\times$ SES studies that use within-family genetic variation on other traits illustrate the residual difficulty: in education, the direction of the gene–SES interaction shifts across developmental stages and is small or null in a Norwegian welfare-state sample using a trio design (Isungset et al., 2022; Houmark et al., 2025), and in smoking the moderation depends on which SES measure is used (Bierut et al., 2023). None of these papers implements the specific contrast required to discriminate direct-effect moderation from family-channel moderation for BMI.

This paper closes the gap by comparing two polygenic indices that differ precisely in their exposure to the indirect and stratification channels—a standard child BMI PGI from population GWAS and a direct genetic effects PGI from family GWAS—estimated on the same children in the same regression. The contrast supplies the identification step that observational studies lack (Biroli et al., 2026; Benjamin et al., 2024). The hypotheses that this contrast can discriminate are formalised in

Section 3.6.

3 Mathematical framework

3.1 The genotype random variable

This section builds the probability model for genotypes that the two GWAS designs of Section 3.2 rely on. The model is developed in two steps. The first step characterises the distribution of the child's allele count conditional on parental genotypes: this is the Mendelian view, used by any analysis that observes both parents. The second step characterises the marginal distribution that remains once parental information is integrated out: this is the Hardy–Weinberg view, used by any analysis that does not condition on parents and that must reckon with population structure and the mating regime instead. The two views together yield four conditional-expectation moments, collected at the end of the section, that are the only probabilistic facts the rest of the paper requires.

3.1.1 Allele counts

Each human cell carries two homologous copies of every autosome, one inherited from each parent. A position at which the population carries more than one nucleotide is a single-nucleotide polymorphism (SNP). Fix such a SNP, indexed by ℓ , designate one of the two observed nucleotides as the reference allele, and let

$$g_{i\ell} = \text{the count of reference-allele copies carried by individual } i \text{ at SNP } \ell. \quad (1)$$

The variable $g_{i\ell}$ takes values in $\{0, 1, 2\}$. To track family members I write $g_{ci\ell}$ for the child of family i and $g_{pi\ell}, g_{mi\ell}$ for the father and mother.

3.1.2 Meiotic transmission

A gamete (sperm, egg) carries one chromosome from each homologous pair. Which of the two parental chromosomes is transmitted is, conditional on the parent's genotype, a fair coin flip (Mendel's first law). Let $T_p^{(i,\ell)} \in \{0, 1\}$ indicate whether the father transmitted the reference allele at SNP ℓ to child i . Conditional on the father's allele count,

$$T_p^{(i,\ell)} \mid g_{pi\ell} \sim \begin{cases} \delta_0 & \text{if } g_{pi\ell} = 0, \\ \text{Bernoulli}(\frac{1}{2}) & \text{if } g_{pi\ell} = 1, \\ \delta_1 & \text{if } g_{pi\ell} = 2, \end{cases} \quad (2)$$

where δ_x is the point mass at x . The three cases collapse to

$$\mathbb{E}[T_p^{(i,\ell)} \mid g_{pi\ell}] = \frac{g_{pi\ell}}{2}, \quad \text{Var}(T_p^{(i,\ell)} \mid g_{pi\ell}) = \frac{1}{4} \mathbf{1}\{g_{pi\ell} = 1\}. \quad (3)$$

An identical statement holds for the maternal indicator $T_m^{(i,\ell)}$. The transmissions of the two parents are independent, since the gametes form in separate organisms.

3.1.3 The child's genotype

The child's allele count is the sum of the two transmitted alleles:

$$g_{cil} = T_p^{(i,\ell)} + T_m^{(i,\ell)}. \quad (4)$$

Taking the conditional expectation of (4) given both parental genotypes and using (3) gives

$$\mathbb{E}[g_{cil} \mid g_{pil}, g_{mil}] = \frac{g_{pil} + g_{mil}}{2}. \quad (5)$$

Define the segregation residual

$$u_{il} = g_{cil} - \frac{g_{pil} + g_{mil}}{2}. \quad (6)$$

Two properties of u_{il} are central to what follows. First, by construction,

$$\mathbb{E}[u_{il} \mid g_{pil}, g_{mil}] = 0. \quad (7)$$

Second, the conditional variance is

$$\text{Var}(u_{il} \mid g_{pil}, g_{mil}) = \frac{1}{4}(\mathbf{1}\{g_{pil} = 1\} + \mathbf{1}\{g_{mil} = 1\}), \quad (8)$$

positive whenever at least one parent is heterozygous.

Equation (7) is the foundation of any identification argument that conditions on parental genotypes. It says that the deviation of a child's genotype from her parents' average is mean zero given the parents themselves; it does not say this deviation is mean zero unconditionally. The deviation is generated by meiotic coin flips that took place before the child was conceived, and any environmental influence that operates through parental traits, household conditions, or shared ancestry can only act on u_{il} through the conditioning variables (g_{pil}, g_{mil}) . Section 3.2.3 returns to this point when it introduces the design that uses this conditioning.

3.1.4 Hardy–Weinberg and ancestry

The previous subsection characterised the conditional distribution of g_{cil} given parental genotypes. The complementary marginal distribution—what g_{cil} looks like once parental information is integrated out—is determined by Hardy–Weinberg equilibrium and the mating regime. Under random (panmictic) mating in an infinite population with no selection or mutation, the two alleles an individual carries are independent draws from a Bernoulli with reference-allele frequency $p_\ell \in [0, 1]$, so

$$g_{il} \sim \text{Binomial}(2, p_\ell), \quad \mathbb{E}[g_{il}] = 2p_\ell, \quad \text{Var}(g_{il}) = 2p_\ell(1 - p_\ell). \quad (9)$$

This is the Hardy–Weinberg distribution. It fails when the sample is drawn from a mixture of subpopulations with different p_ℓ , when partners are correlated on heritable traits, or under strong selection. The first two failures matter for both GWAS designs treated below.

Population structure and strata. Let Z_i denote the subpopulation membership of individual i . I treat Z_i as a discrete-valued unobservable: the analyst cannot read it off the data, and no regression on observable data can place Z_i directly on its right-hand side. A stratum is a level of Z_i . The

modelling assumption is that within each stratum the reference-allele frequency is a single number $p_\ell(Z_i)$ and the two alleles an individual carries are conditionally independent, so Hardy–Weinberg holds within stratum:

$$g_{i\ell} \mid Z_i \sim \text{Binomial}(2, p_\ell(Z_i)), \quad \mathbb{E}[g_{i\ell} \mid Z_i] = 2p_\ell(Z_i). \quad (10)$$

The function $p_\ell(\cdot)$ summarises the entire confounding pathway through which population stratification enters GWAS estimates, and also supplies the input to the Mendelian imputation procedure (Section 3.2.3) used to recover unobserved parental genotypes. The resolution at which Z_i is defined is a modelling choice. Coarse strata leave residual allele-frequency variation within levels that (10) treats as zero; finer strata exhaust this variation but contain fewer samples per level.

Equation (10) applies to any individual drawn from the population. Both parents of child i satisfy it with the same function:

$$\mathbb{E}[g_{pi\ell} \mid Z_i] = \mathbb{E}[g_{mi\ell} \mid Z_i] = 2p_\ell(Z_i). \quad (11)$$

The child’s distribution given Z_i alone is identical, $\mathbb{E}[g_{ci\ell} \mid Z_i] = 2p_\ell(Z_i)$, either directly from within-stratum HWE applied to the child or by combining (5) with (11) through the tower property. The child’s allele count and each parent’s allele count carry the same stratum-level information about Z_i .

Mating regime. Random mating means that, conditional on Z_i , partner selection is independent of genotype:

$$g_{pi\ell} \perp g_{mi\ell} \mid Z_i. \quad (12)$$

Phenotypically assortative mating denotes the failure of (12). Partners match on heritable traits, so their allele counts at any SNP that affects those traits become correlated within strata. The same matching also induces correlations between allele counts at distinct SNPs that are physically unlinked: trait-increasing alleles at SNP ℓ and at any other SNP ℓ' that influences the same trait accumulate together in mate-pairs and propagate together to the next generation. This is the channel Young (2023) characterises as genome-wide cross-locus linkage disequilibrium under assortative mating. It is the residual confounding that conditioning on parental genotypes at SNP ℓ alone cannot absorb; Section 3.3.2 returns to it.

Four conditional moments. The remainder of the paper appeals to four conditional-expectation facts from this section, and only four:

1. The Mendelian mean, $\mathbb{E}[g_{ci\ell} \mid g_{pi\ell}, g_{mi\ell}] = (g_{pi\ell} + g_{mi\ell})/2$, equation (5), with its segregation residual $u_{i\ell}$ satisfying (7).
2. The child’s within-stratum mean, $\mathbb{E}[g_{ci\ell} \mid Z_i] = 2p_\ell(Z_i)$, equation (10).
3. The parents’ within-stratum mean, $\mathbb{E}[g_{pi\ell} \mid Z_i] = \mathbb{E}[g_{mi\ell} \mid Z_i] = 2p_\ell(Z_i)$, equation (11).
4. The mating-regime conditional independence, $g_{pi\ell} \perp g_{mi\ell} \mid Z_i$ under random mating, equation (12); the violation under assortative mating, with its cross-locus consequence between unlinked SNPs.

The four moments support two complementary identification strategies, developed in Section 3.2. A regression on the child’s allele count alone—without parental conditioning—relies on (2) and (3) to characterise how residual ancestry shifts the child’s marginal mean. A regression that adds parental genotypes as controls uses (1) together with (3): the Mendelian mean (1) gives the segregation residual orthogonal to parents, and (3) makes parental allele counts informative about $p_\ell(Z_i)$ so that conditioning on them absorbs stratum-level variation in the child’s genotype. The residual confounding of the parent-conditioning strategy under assortative mating is the failure of (4).

3.2 Two GWAS designs

Two designs answer the question of how a single SNP enters the outcome. Population GWAS regresses the phenotype on the child’s allele count alone, marginalising over parental genotypes; family GWAS adds parental genotypes as controls. The two designs target the same parameter—the child’s own direct genetic effect—but reach it under different identifying assumptions, with different residual biases. This section writes both regressions, derives the bias decomposition that the population-GWAS estimand inherits when the parental and ancestry channels are non-zero, and tables the two assumption sets side by side.

3.2.1 The outcome model

I model how SNP ℓ enters a continuous outcome y_i (BMI, in this application). Any variable that is correlated with the child’s allele count g_{cil} and also affects y_i creates a pathway through which g_{cil} would appear to matter for y_i beyond its own biological role. Each such pathway has to appear as a separate term on the right-hand side of the outcome equation; only quantities independent of every listed term are absorbed into the residual. Three pathways pass this test. The child’s own allele count enters through her own biology. The parents’ allele counts g_{pil}, g_{mil} are mechanically correlated with g_{cil} through Mendelian transmission, equation (5), and reach y_i through the household environment that parental genotypes shape. The child’s latent stratum membership Z_i shifts her allele count through $p_\ell(Z_i)$, equation (10), and shifts y_i through stratum-specific environments. Writing the four contributions out gives the outcome equation

$$y_i = \alpha + \beta_\ell^{\text{dir}} g_{cil} + \eta_\ell^p g_{pil} + \eta_\ell^m g_{mil} + \eta_\ell^Z Z_i + u_i. \quad (13)$$

The four structural coefficients name the four pathways. The direct genetic effect β_ℓ^{dir} is the per-allele change in y_i generated through the child’s own biology, holding the other three channels fixed. The parental indirect effects η_ℓ^p, η_ℓ^m are the parts of y_i generated by each parent’s allele count, net of the child’s own, through the household environment that parental genotypes shape. The stratification coefficient η_ℓ^Z is the part of y_i generated by stratum membership through stratum-specific environments that do not act through parental genotype.

The structural content of the parental indirect terms is made explicit by one short substitution. Suppose

$$E_i = \alpha_p g_{pil} + \alpha_m g_{mil} + U_i^E, \quad y_i = \beta_\ell^{\text{dir}} g_{cil} + \gamma E_i + U_i^y. \quad (14)$$

Substituting the first equation of (14) into the second reproduces the parental terms of (13) with $\eta_\ell^p = \gamma \alpha_p$ and $\eta_\ell^m = \gamma \alpha_m$, and composite residual $u_i = \gamma U_i^E + U_i^y$. The substitution covers the parental indirect channel only. The stratification term $\eta_\ell^Z Z_i$ remains a separate channel: Z_i acts on y_i through stratum-specific environments that are not built from parents’ allele counts at SNP ℓ .

The residual u_i collects every source of variation in y_i that (13) treats as orthogonal to the four listed channels. It contains idiosyncratic environmental influences and background genetic variation at all other SNPs of the genome. The latter sits passively in u_i under random mating; it becomes a binding confounder when assortative mating induces cross-locus correlations between SNP ℓ and other SNPs in the residual (Section 3.3.2).

I maintain three assumptions on (13) throughout the paper. (i) The relation is additive in g_{cil} : no SNP-by-SNP epistasis. (ii) The structural coefficients $(\beta_\ell^{\text{dir}}, \eta_\ell^p, \eta_\ell^m, \eta_\ell^Z)$ are constant across individuals at a given SNP. (iii) The residual u_i is mean-independent of $(g_{cil}, g_{pil}, g_{mil}, Z_i)$. These are model-level assumptions: they make (13) a valid linear structural model with well-defined coefficients. Identifying any one coefficient from data requires further design-specific assumptions, which I introduce design by design below. Section 3.3 returns to how plausible (i)–(iii) are for BMI.

3.2.2 Population GWAS

Population GWAS regresses the phenotype on the child’s allele count alone, with a vector of pre-determined controls $\mathbf{c}_i$ containing sex, age, and the leading genetic principal components:

$$y_i = \alpha + \beta_\ell g_{cil} + \boldsymbol{\gamma}' \mathbf{c}_i + \varepsilon_i. \quad (15)$$

The regression is run separately for every SNP ℓ . The latent stratum membership Z_i does not appear on the right-hand side: it is unobservable by construction (Section 3.1.4). The leading principal components inside $\mathbf{c}_i$ act as observable proxies for Z_i , absorbing whatever stratum-level variation they can project onto; variation in Z_i orthogonal to the leading PCs is swept into the residual ε_i . The OLS estimate $\hat{\beta}_\ell$ is interpreted by standard practice as an estimate of the direct genetic effect β_ℓ^{dir} from (13).

The regression omits the parental allele counts (g_{pil}, g_{mil}) and the latent Z_i entirely. Each is correlated with g_{cil} through a conditional moment from Section 3.1.4 and each affects y_i by (13). Write $\mathbf{W}_i = (g_{pil}, g_{mil}, Z_i)'$ for the column vector of omitted variables and $\boldsymbol{\eta}_\ell = (\eta_\ell^p, \eta_\ell^m, \eta_\ell^Z)'$ for the corresponding structural coefficients. The OLS estimator $\hat{\beta}_\ell$ has probability limit

$$\beta_\ell^* = \beta_\ell^{\text{dir}} + \boldsymbol{\eta}_\ell' \boldsymbol{\pi}_\ell, \quad (16)$$

where $\boldsymbol{\pi}_\ell$ is the vector of coefficients from regressing $\mathbf{W}_i$ on g_{cil} , all moments residualised on $\mathbf{c}_i$. Each entry of $\boldsymbol{\pi}_\ell$ is a partial regression coefficient computed jointly with the others, because (g_{pil}, g_{mil}) and Z_i are themselves correlated: parents are drawn from stratum Z_i under (11), so their allele counts carry information about $p_\ell(Z_i)$.

The joint bias $\boldsymbol{\eta}_\ell' \boldsymbol{\pi}_\ell$ groups into two labelled components,

$$\beta_\ell^* = \beta_\ell^{\text{dir}} + \beta_\ell^{\text{ind}} + \beta_\ell^{\text{conf}}, \quad (17)$$

where β_ℓ^{ind} is the indirect-effect contribution from the parental block (g_{pil}, g_{mil}) and β_ℓ^{conf} is the stratification contribution from Z_i after the parental block has been partialled out. Under (11) the parental block and Z_i share variance, so the split between β_ℓ^{ind} and β_ℓ^{conf} depends on partialling order. I partial parents out first: under this order β_ℓ^{ind} is the bias that the family GWAS design of Section 3.2.3 absorbs by conditioning on parental genotypes, and β_ℓ^{conf} is the residual stratification the design does not absorb. Population GWAS recovers the direct effect when both labelled contributions vanish.

Collect the population-GWAS identifying conditions under labels, for later cross-reference with family GWAS in Section 3.2.3. The OLS estimator $\hat{\beta}_\ell$ from (15) is consistent for β_ℓ^{dir} when (PG-A) the standing assumptions (i)–(iii) on (13) hold; (PG-B) $\eta_\ell^p = \eta_\ell^m = 0$, so parents’ alleles do not affect the child’s outcome except through the child’s own alleles; and (PG-C) $\mathbf{c}_i$ is sufficient to render Z_i conditionally orthogonal to $g_{c\ell}$ at SNP ℓ . The next subsection writes down a design that relaxes (PG-B) and weakens (PG-C), at the cost of additional assumptions.

3.2.3 Family GWAS

Family GWAS adds the parents’ allele counts to the regression:

$$y_i = \delta_\ell g_{c\ell} + \alpha_{p\ell} g_{p\ell} + \alpha_{m\ell} g_{m\ell} + \gamma' \mathbf{c}_i + \varepsilon_i. \quad (18)$$

As in (15), $\mathbf{c}_i$ contains sex, age, and the leading principal components; the latent Z_i remains unobservable and is again proxied through the PCs. The coefficient of interest is δ_ℓ . The parental slopes $\alpha_{p\ell}, \alpha_{m\ell}$ are auxiliary: they absorb the parental indirect channel but are not the target of inference.

Why this regression isolates the direct effect. By Frisch–Waugh–Lovell, the OLS estimate of δ_ℓ from (18) equals the slope from regressing y_i on the residual of $g_{c\ell}$ after $(g_{p\ell}, g_{m\ell}, \mathbf{c}_i)$ have been partialled out. The Mendelian conditional mean (5) states that the linear projection of $g_{c\ell}$ on $(g_{p\ell}, g_{m\ell})$ is exactly $(g_{p\ell} + g_{m\ell})/2$, so this residual coincides with the segregation residual $u_{i\ell}$ of (6) up to the orthogonalisation against $\mathbf{c}_i$. The segregation residual has mean zero given parental genotypes (equation 7) and is generated by meiotic coin flips that occurred before the child was conceived. Under the maintained assumption that no other determinant of y_i is correlated with $u_{i\ell}$ conditional on $(g_{p\ell}, g_{m\ell}, \mathbf{c}_i)$, the OLS estimator is consistent for $\delta_\ell = \beta_\ell^{\text{dir}}$.

The stratification channel is partially absorbed by the same conditioning. Under (10)–(11), the parental block $(g_{p\ell}, g_{m\ell})$ and the child’s allele count $g_{c\ell}$ have the same stratum-conditional mean, namely $2p_\ell(Z_i)$. Conditioning on the parental block therefore removes the part of $g_{c\ell}$ that comes from stratum-level variation in p_ℓ . What survives are pathways through which Z_i correlates with $g_{c\ell}$ but that are not encoded by parents’ allele counts at SNP ℓ . Two such pathways matter for BMI. First, residual structure within strata: when (10) holds only at the resolution of the stratum definition, p_ℓ varies within strata in ways the parental SNP- ℓ genotypes cannot project onto. Second, cross-locus correlations from assortative mating: the residual u_i contains background genetic variation at all other SNPs of the genome, and the failure of (12) across loci makes trait-increasing alleles at SNPs other than ℓ correlated with $g_{c\ell}$ even after the parents’ SNP- ℓ genotypes have been controlled for. Section 3.3.2 returns to both channels.

Mendelian imputation when one parent is missing. The trio regression (18) requires both parental genotypes, but in practice many children have only one parent genotyped, typically the mother. The design can be extended by imputing the missing parent (Young et al., 2022). For a child with observed mother but unobserved father, the imputed paternal allele count is

$$\hat{g}_{p\ell} = \mathbb{E}[g_{p\ell} \mid g_{c\ell}, g_{m\ell}, p_\ell(Z_i)], \quad (19)$$

which has a closed form under Mendel’s law (§3.1.2) and within-stratum HWE (10): the father’s transmitted allele is identified from $g_{c\ell}$ and $g_{m\ell}$, and the non-transmitted allele is a draw from

Bernoulli($p_\ell(Z_i)$). Substituting $\hat{g}_{pil}$ for g_{pil} in (18) preserves the estimand: $\hat{g}_{pil}$ is the conditional mean of g_{pil} , so the residual $\nu_{il} = g_{pil} - \hat{g}_{pil}$ is uncorrelated with $\hat{g}_{pil}$ by the projection identity, and the consistency argument of the trio regression carries through with inflated standard errors. The imputation extension adds one identifying assumption to the trio case: that (10) provides the correct allele-frequency function $p_\ell(\cdot)$ at the chosen stratum resolution.

Identifying assumptions for family GWAS. Beyond the standing assumptions (i)–(iii) on (13), the trio regression (18) recovers $\delta_\ell = \beta_\ell^{\text{dir}}$ under four further conditions. (FG-A) Mendelian transmission delivers the fair-coin distribution (2); equivalently, the Mendelian mean (5) holds exactly. (FG-B) The reduced-form environment depends only on the focal child’s parents’ allele counts at SNP ℓ , not on siblings, grandparents, or extra-household relatives. This restricts the source of indirect channels to the parental layer at SNP ℓ . (FG-C) The residual u_i in (13) is mean-independent of g_{cil} conditional on $(g_{pil}, g_{mil}, \mathbf{c}_i)$. This is the family-GWAS analogue of (PG-C). It fails when assortative mating violates (12) across loci: trait-increasing alleles at SNPs other than ℓ are absorbed into u_i as background genetic variation, and the cross-locus correlations induced by mate-matching make those alleles correlated with g_{cil} even after parents at SNP ℓ have been controlled for. Young (2023) develops corrections at the genome-aggregation level—for heritability estimation and for PGI-based estimators of indirect genetic effects—under assortative mating; a closed-form correction for the single-SNP coefficient δ_ℓ in (18) is not provided by the existing literature. (FG-D) For the imputation extension (19), (10) delivers the correct allele-frequency function $p_\ell(\cdot)$ at the chosen stratum resolution.

3.2.4 The two designs, compared

Table 1 places the assumption sets of the two designs side by side. The family-GWAS design does not strictly nest the population-GWAS design: it relaxes (PG-B) and weakens (PG-C), but it adds (FG-A), (FG-B), (FG-C), and (FG-D). Each substitution carries an empirical risk specific to the assumption being added. The trade-off is the subject of Section 3.3.

3.3 What each design fails to deliver

Each design carries a distinctive set of identifying failures. Population GWAS fails through the indirect-effect and stratification components of equation (17): the population-GWAS coefficient is contaminated whenever parental allele counts or residual ancestry act on the outcome through pathways that the controls do not absorb. Family GWAS substitutes one assumption set for another and inherits its own failure modes: a smaller discovery sample that erodes the DGE PGI’s precision, channels that operate outside the immediate parents (siblings, grandparents), assortative mating’s genome-wide LD across loci, the within-stratum HWE assumption that underwrites Mendelian imputation, and the constant-coefficient commitment that the gene-environment-interaction question puts under particular strain. This section walks through both sets and, where the literature allows, bounds the empirical bite of each for childhood BMI.

3.3.1 Failures of population GWAS

Equation (17) catalogues the three components of the population-GWAS estimand. Two of them, the indirect and the stratification residuals, are biases relative to the conceptual target β_ℓ^{dir} .

Table 1: Identifying assumptions, side by side

Assumption	Population GWAS	Family GWAS
Additive, constant-coefficient $g \rightarrow y$	Required (standing)	Required (standing)
$\eta_\ell^p = \eta_\ell^m = 0$ (no parental indirect effect)	Required (PG-B)	Not required
Leading PCs sufficient for Z_i	Required (PG-C)	Weakened: parents' allele counts absorb the stratum-level component of g_{cil} through (11)
Mendelian fair-coin transmission	Not invoked	Required (FG-A)
Reduced-form environment depends only on parents' allele counts at SNP ℓ (no sibling, grandparental, or extra-household indirect channels)	Not relevant	Required (FG-B)
No cross-locus correlation between g_{cil} and u_i beyond what parental genotypes at SNP ℓ encode	Not relevant	Required (FG-C)
Within-stratum HWE for $p_\ell(\cdot)$	Not required	Required for imputation of missing parents (FG-D)

Notes: "Required" means the assumption must hold for the design to recover β_ℓ^{dir} . The family-GWAS design does not strictly weaken the population-GWAS assumption set; it substitutes one set of assumptions for another. Section 3.3 examines each substitution.

Indirect effects (genetic nurture). Parents’ allele counts are correlated with the child’s by Mendel’s law and influence the child’s outcome through the household environment they generate. Kong et al. (2018) establish the phenomenon empirically by showing that polygenic scores built from parents’ non-transmitted alleles still predict child outcomes through environmental pathways. Howe et al. (2022) quantify the gap between within-sibship and population-based GWAS estimates across 25 traits. For educational attainment, within-sibship estimates are 47% smaller than population estimates (95% CI 41% to 52%). For BMI, in contrast, the gap is not statistically distinguishable from zero: the within-sibship-to-population Mendelian-randomisation slope is -5% (95% CI -14% to 4%). The family-GWAS meta-analysis of Tan et al. (2026) confirms the same pattern at the family-GWAS level. The SNP heritability from direct genetic effects, 0.212 (S.E. 0.013), is statistically indistinguishable from the SNP heritability estimated from population effects, 0.216 (S.E. 0.012). The genome-wide variance share of population effects uncorrelated with direct effects is only 10.2% (S.E. 1.0%). The indirect-effect contribution to the standard BMI PGI is therefore small on average. Any standard-versus-DGE difference detected in the gene-environment-interaction coefficient must operate through a channel that the average comparison does not detect—the empirical question of Section 3.5.

Population stratification. Subpopulations differ in allele frequencies and in environments. The leading K principal components of the genotype matrix absorb a rank- K approximation to the ancestry signal, but residual structure survives in higher singular directions (Novembre et al., 2008), in identity-by-descent segments shared by recent ancestors that do not align with the population-level allele-frequency differences (Yang et al., 2014), and in long-range linkage disequilibrium induced by phenotypically assortative mating (Yengo et al., 2018a; Howe et al., 2022). The residual stratification bias β_ℓ^{conf} is plausibly small in a single continental-ancestry sample with ten PCs included, but it is not zero and its magnitude is harder to bound empirically than the indirect contribution.

3.3.2 Failures of family GWAS

The family-GWAS design substitutes one assumption set for another. Each new assumption introduces its own failure mode.

Discovery-sample size penalty. Family-GWAS discovery samples are smaller than population-GWAS discovery samples by roughly an order of magnitude—effective $N \approx 82,000$ for the Tan et al. (2026) family-GWAS BMI summary statistics, against $N \approx 700,000$ for the Yengo et al. (2018b) population-GWAS meta-analysis. Asymptotically the DGE PGI is unbiased for the latent direct-effect score and the Standard PGI is biased for it; in finite samples the two indices load on the direct channel with different weights $w_{\text{Std}}^{\text{dir}}$ and $w_{\text{DGE}}^{\text{dir}}$ (23)–(24). Because the two indices load on the direct channel with different weights, their SES interactions are not equal even under pure H1; they stand in a fixed ratio set by the weight gap. The H1/H2 contrast is therefore a test of that ratio, not of equality; Section 7.4 carries this through to the empirical bite for childhood BMI.

Channels outside the immediate parents. The reduced-form environment in (14) writes household conditions as a function of parental genotypes only. Other relatives’ genotypes can shape the focal child’s environment through channels the design does not absorb. Siblings can act through shared meals, peer interactions, or differential parental attention. A grandparent can act through the parent’s upbringing in a way that the parent’s own genotype only partially

captures. Extra-household relatives—uncles, cousins—whose genotypes are correlated with the child’s through pedigree and whose behaviour affects the child can act through any of these routes. The family-GWAS design treats each of these channels as part of the residual. The existing within-sibship and family-based GWAS literature focuses on the parental layer and does not separately quantify the sibling, grandparental, or extra-household components.

Assortative mating beyond the parental layer. Phenotypically assortative mating—partners matched on heritable traits—violates the mating-regime conditional independence (12) both within strata at SNP ℓ and across loci genome-wide. The within-locus violation is observable: both parents of the focal child enter (18) as regressors, and any correlation between g_{pil} and g_{mil} within strata is absorbed in the partialling. The cross-locus violation is not. Trait-increasing alleles at SNPs other than ℓ sit in the residual u_i as background genetic variation, and the genome-wide LD induced by partner matching at those other loci leaves u_i correlated with g_{cil} even after conditioning on (g_{pil}, g_{mil}) . This is the failure of (FG-C). Young (2023) establishes the bias for heritability estimation and for PGI-based two-generation estimators of indirect genetic effects under assortative mating, and proposes corrections at the genome-aggregation level; the existing literature does not give a closed-form expression for the residual bias on the single-SNP coefficient δ_ℓ in (18). For BMI specifically, the meta-analytic mate-pairs analysis of Yengo et al. (2018a) reports $\theta = 3.2\%$ for height and $\theta = 2.7\%$ for educational attainment, but the GWAS sample size available for BMI was not large enough to identify the corresponding signal for BMI. The assortative-mating bias on the family-GWAS direct-effect coefficient for BMI is therefore plausibly small. I do not implement a correction; the residual bias is plausibly small but cannot be ruled out.

The HWE-within-strata assumption. The imputation (19) requires the allele-frequency function $p_\ell(\cdot)$ as input and assumes within-stratum HWE: within each stratum, the two alleles an individual carries are independent draws. Violations occur under finer population structure than the stratum partition captures and under recent admixture. A systematic departure from (10) would bias the imputed paternal genotype and therefore $\hat{\delta}_\ell$.

The stability of the direct effect across environments. Equation (13) treats β_ℓ^{dir} as a constant scalar. This is the deepest commitment of the model and the one most directly in tension with the gene-environment-interaction question the paper studies. A gene-environment interaction in the strict biological sense—where the per-allele effect on metabolism, appetite, or activity depends on the surrounding environment—makes β_ℓ^{dir} itself a function of SES. The family-GWAS design then identifies a mean of β_ℓ^{dir} over the joint distribution of SES and other moderators in the discovery sample. The SES-specific parameter that the interaction model targets is a different object. The DGE PGI built from these mean-weights is an estimator of a population-mean parameter; its interaction with an environmental variable is informative about the SES gradient in that population mean, but it is not in general equal to a structural direct gene-by-environment effect.

I adopt the constant-coefficient specification (13) as a maintained working model; the interpretive implication for $\hat{\beta}_{GS}^{\text{DGE}}$ is taken up in Section 7.4.

3.4 The polygenic index

The single-SNP analysis of Sections 3.2.2 and 3.2.3 establishes what a population-GWAS or family-GWAS coefficient identifies. The gene-environment-interaction question of this paper, however, requires individual-level predictors because single SNPs explain only a small fraction of BMI variance. This section aggregates the SNP-level estimates into a polygenic index (PGI) for each design, transfers the bias decomposition of equation (17) from the SNP level to the score level, and notes the finite-sample noise the family-GWAS PGI inherits from its smaller discovery sample.

The polygenic index for child i is the weighted sum of SNP-level allele counts using GWAS-estimated weights $\hat{\beta}_\ell$:

$$\widehat{\text{PGI}}_i = \sum_{\ell=1}^L \hat{\beta}_\ell g_{ci\ell}. \quad (20)$$

Substituting different SNP weights produces different scores on the same genotype data. I construct two indices for each child in the MCS. The Standard PGI uses the population-GWAS weights of Yengo et al. (2018b), shrunk by LDpred (Vilhjálmsdóttir et al., 2015). The DGE PGI uses the family-GWAS weights of Tan et al. (2026), scored on each family by `snipar` (Young et al., 2022) with Mendelian-imputed allele counts substituted for missing parents.

The score-level estimands inherit the SNP-level estimands by linearity. The decomposition of the population-GWAS coefficient (17) aggregates into

$$\widehat{\text{PGI}}_i^{\text{Std}} = \widehat{\text{PGI}}_i^{\text{dir}} + \widehat{\text{PGI}}_i^{\text{ind}} + \widehat{\text{PGI}}_i^{\text{conf}}, \quad (21)$$

each component evaluated on the child's own genotypes with a component-specific weight vector. The family-GWAS weights target β_ℓ^{dir} asymptotically, so

$$\widehat{\text{PGI}}_i^{\text{DGE}} \xrightarrow{p} \widehat{\text{PGI}}_i^{\text{dir}} \text{ as the family-GWAS discovery sample grows without bound.}$$

In this application the discovery sample is finite at $N \approx 82,000$, so the realised DGE PGI is a noisy version of the latent direct-effect score, with the noise governed by the family-GWAS-specific R^2 -to-heritability ratio (Becker et al., 2021). The Standard PGI is, by contrast, biased even asymptotically: the decomposition (21) does not collapse to a single channel as the discovery sample grows.

3.5 The gene-environment estimating equation

The gene-environment interaction is estimated by adding a $\text{PGI} \times \text{SES}$ term to the BMI regression. Which PGI enters the equation determines what the interaction coefficient estimates. This section writes the estimating equation, propagates the PGI bias decomposition of equation (21) into the interaction coefficient, and shows that the difference between the Standard and DGE interactions identifies the family-channel contribution provided the direct-channel weights of the two PGIs are similar enough that their gap does not swamp the family-channel signal.

The estimating equation for the interaction of a polygenic index with socioeconomic status is

$$\text{BMI}_{it} = \alpha + \beta_G \widehat{\text{PGI}}_i + \beta_S \text{SES}_i + \beta_{GS} (\widehat{\text{PGI}}_i \times \text{SES}_i) + \delta_t + \mathbf{X}'_i \boldsymbol{\gamma} + \varepsilon_{it}, \quad (22)$$

estimated as a child-wave panel with wave fixed effects δ_t and controls $\mathbf{X}_i$ (sex, cohort indicators, ten PCs). The coefficient β_{GS} is the gene-environment interaction.

What β_{GS} estimates depends on which PGI enters (22). Substituting (21) into (22) and partialling parents out first, as in §3.2.2,

$$\beta_{GS}^{\text{Std}} = w_{\text{Std}}^{\text{dir}} \beta_{GS}^{\text{dir}} + w_{\text{Std}}^{\text{ind}} \beta_{GS}^{\text{ind}} + w_{\text{Std}}^{\text{conf}} \beta_{GS}^{\text{conf}}, \quad (23)$$

where each β_{GS}^c is the population interaction coefficient that would be recovered if the channel-specific score were observed separately, and each w_{Std}^c is the partial regression coefficient of $\widehat{\text{PGI}}_i^c \times \text{SES}_i$ on $\widehat{\text{PGI}}_i^{\text{Std}} \times \text{SES}_i$ in the analysis sample, conditional on the controls. The partialling order matches the one used for (17). A non-zero β_{GS}^{Std} is consistent with non-zero β_{GS}^{dir} , with non-zero β_{GS}^{ind} , or with non-zero β_{GS}^{conf} , in any combination.

When the DGE PGI replaces the Standard PGI in (22), the indirect and stratification weights vanish asymptotically and the estimand reduces to

$$\beta_{GS}^{\text{DGE}} = w_{\text{DGE}}^{\text{dir}} \beta_{GS}^{\text{dir}}. \quad (24)$$

The weight $w_{\text{DGE}}^{\text{dir}}$ is below one in finite samples; the slack is carried by the discovery-stage measurement error of the DGE PGI rather than by the indirect or stratification channels.

The difference between the two estimands isolates the family-channel contribution to the standard interaction:

$$\beta_{GS}^{\text{Std}} - \beta_{GS}^{\text{DGE}} = (w_{\text{Std}}^{\text{dir}} - w_{\text{DGE}}^{\text{dir}}) \beta_{GS}^{\text{dir}} + w_{\text{Std}}^{\text{ind}} \beta_{GS}^{\text{ind}} + w_{\text{Std}}^{\text{conf}} \beta_{GS}^{\text{conf}}. \quad (25)$$

Under the maintained assumption that residual stratification is small after the PC adjustments and that the direct-channel weights are similar across the two scores up to the precision penalty, the second term of (25) is the dominant component of the contrast. The design then recovers the family-channel contribution; whether that maintained assumption holds for the current DGE PGI is an empirical question, taken up in §6.3 and §7.4. I implement the contrast as a restriction test on a single regression containing both indices and both interactions; the formal statement and the equation defining the joint regression are in Section 5.5.

3.6 Hypotheses

The Standard PGI $\times$ SES interaction for childhood BMI is reliably negative in European-ancestry samples: the standard PGI's slope on BMI is steeper at lower SES than at higher SES. The framework of Sections 3.2–3.5 maps three underlying structural interactions onto this observable—a direct-effect interaction β_{GS}^{dir} , an indirect-channel interaction β_{GS}^{ind} , and a stratification-channel interaction β_{GS}^{conf} . This section labels three hypotheses about which of those parameters is non-zero, derives the observable predictions each generates for the Standard and DGE interactions, and operationalises the H1/H2 contrast as a restriction test.

H1: direct-effect moderation. The SES gradient operates through the child's own biological direct effect:

$$\beta_{GS}^{\text{dir}} < 0, \quad \beta_{GS}^{\text{ind}} = 0, \quad \beta_{GS}^{\text{conf}} = 0.$$

The observable estimands then satisfy $\beta_{GS}^{\text{Std}} = w_{\text{Std}}^{\text{dir}} \beta_{GS}^{\text{dir}}$ and $\beta_{GS}^{\text{DGE}} = w_{\text{DGE}}^{\text{dir}} \beta_{GS}^{\text{dir}}$. Both are negative. Their ratio equals $w_{\text{Std}}^{\text{dir}}/w_{\text{DGE}}^{\text{dir}}$, which is not in general one: in finite samples the two PGIs load on

the direct channel with different weights determined by their respective discovery sample sizes and construction methods.

H2: family-channel moderation. The SES gradient operates through parental and household-level pathways embedded in the Standard PGI:

$$\beta_{GS}^{\text{dir}} = 0, \quad \beta_{GS}^{\text{ind}} < 0, \quad \beta_{GS}^{\text{conf}} \approx 0.$$

The observable estimands then satisfy $\beta_{GS}^{\text{Std}} = w_{\text{Std}}^{\text{ind}} \beta_{GS}^{\text{ind}} < 0$ and $\beta_{GS}^{\text{DGE}} = 0$.

Mixture. Both channels operate, so the observable estimands include nonzero contributions from each. Define the family-channel share $s_{\text{fam}} = w_{\text{Std}}^{\text{ind}} \beta_{GS}^{\text{ind}} / \beta_{GS}^{\text{Std}}$ under the simplifying restriction $\beta_{GS}^{\text{conf}} = 0$; $s_{\text{fam}} = 0$ recovers H1 and $s_{\text{fam}} = 1$ recovers H2. Table 2 summarises the three cases.

Table 2: Observable predictions under each hypothesis

Hypothesis	Standard PGI $\times$ SES	DGE PGI $\times$ SES
H1: direct-effect moderation	Negative; $= w_{\text{Std}}^{\text{dir}} \beta_{GS}^{\text{dir}}$.	Negative; $= w_{\text{DGE}}^{\text{dir}} \beta_{GS}^{\text{dir}}$. Same sign as Standard; magnitude scaled by the ratio of direct-channel weights, which is bounded above by one and shrinks with finite family-GWAS discovery.
H2: family-channel moderation	Negative; $= w_{\text{Std}}^{\text{ind}} \beta_{GS}^{\text{ind}}$.	Zero (asymptotically).
H1 + H2 mixture	Negative; both direct- and indirect-channel contributions.	Negative but smaller in magnitude than the Standard interaction; the difference $\beta_{GS}^{\text{Std}} - \beta_{GS}^{\text{DGE}}$ combines (i) the family-channel contribution $w_{\text{Std}}^{\text{ind}} \beta_{GS}^{\text{ind}}$ and (ii) the direct-channel weight gap $(w_{\text{Std}}^{\text{dir}} - w_{\text{DGE}}^{\text{dir}}) \beta_{GS}^{\text{dir}}$.

Notes: All three hypotheses generate a negative Standard interaction. The DGE PGI distinguishes them only conditional on what is known about the direct-channel weight gap $w_{\text{Std}}^{\text{dir}} - w_{\text{DGE}}^{\text{dir}}$. Under H1, this gap drives the entire $\beta_{GS}^{\text{Std}} - \beta_{GS}^{\text{DGE}}$ difference; under H2, the difference is driven by the indirect channel; under the mixture both contribute. The contrast is implemented as a Wald restriction test on $H_0 : \beta_{GS}^{\text{Std}} = \beta_{GS}^{\text{DGE}}$ in a single regression containing both PGIs and both interactions (§5.5). The divergence between the two interactions is consistent with H2, with the mixture, **or** with H1 when $w_{\text{Std}}^{\text{dir}} \neq w_{\text{DGE}}^{\text{dir}}$; the divergence alone does not discriminate H1-with-weight-gap from H2. Discrimination requires comparing the divergence against the ratio pure H1 implies, set by the PGIs' relative predictive accuracy on BMI.

4 Data

The data come from the UK Millennium Cohort Study and, for each genotyped child, comprise repeated BMI measurements, baseline household income, two trait-matched polygenic indices, and

standard demographic and ancestry controls. This section describes the sample, the outcome and PGI construction at a high level (the variant-level pipeline is in Appendix 7.5), the socioeconomic-status measure used as the moderating environment, and the two further outcomes—smoking and child behavioural difficulties—that serve the mechanism-heterogeneity analysis of Section 6.6.

4.1 Sample and outcome

The Millennium Cohort Study (MCS) is a nationally representative UK birth cohort of approximately 19,000 children born in 2000–2002, with repeated measures collected at five sweeps (ages 5, 7, 11, 14, and 17) and a subset of children and parents genotyped. The genotyped sample is restricted to children of European ancestry, consistent with the discovery ancestry of both the [Yengo et al. \(2018b\)](#) and [Tan et al. \(2026\)](#) GWAS summary statistics from which the polygenic indices are constructed.

The outcome is body mass index, defined as weight in kilograms divided by height in metres squared. The unit of observation in panel specifications is a child–wave pair (i, t) . The estimation sample is the subset of genotyped children with non-missing BMI, baseline SES, and control variables, restricted to a common sample across specifications within each table; descriptive statistics appear in Table 3. The sample contains 5,446 children from 5,411 families, with 35 co-enrolled sibling pairs (1.3% of children); sibling-mediated channels between the focal child and other genotyped children in the sample are therefore mechanically small.

4.2 Polygenic indices

Two child-level polygenic indices for BMI are constructed from publicly available GWAS summary statistics applied to the MCS genotype data; the variant-level pipeline is in Appendix 7.5.

The **Standard BMI PGI** uses the population-GWAS weights of [Yengo et al. \(2018b\)](#) (discovery $N \approx 700,000$, European ancestry), shrunk by LDpred ([Vilhjálmsdóttir et al., 2015](#)) under the infinitesimal prior. In the MCS estimation sample it explains 4.9% of cross-sectional BMI variance ($\hat{\beta}_G = 0.92 \text{ kg/m}^2 \text{ per SD}$; $p < 10^{-140}$).

The **direct genetic effects (DGE) BMI PGI** uses the family-GWAS weights of [Tan et al. \(2026\)](#) (effective $N \approx 82,000$; direct-effect SNP heritability $h^2 = 0.21$, SE 0.01), scored by snipar ([Young et al., 2022](#)) with Mendelian-imputed allele counts substituted for missing parents (§3.2.3, equation 19). In the MCS estimation sample it explains 0.04% of cross-sectional BMI variance ($\hat{\beta}_G = 0.09 \text{ kg/m}^2 \text{ per SD}$; $p = 0.02$).

PGIs are standardised to mean zero and unit variance in the full genotyped MCS sample. Maternal and paternal BMI PGIs are produced jointly with the child index and enter extended specifications as parental controls.

4.3 Socioeconomic status

SES is measured by equivalised household income at baseline using the modified OECD scale (weight 1.0 for the first adult, 0.5 for each additional member aged 14+, 0.3 for each child under 14). The primary specification groups income into quintiles, coded 1 (lowest) to 5 (highest) and centred at the middle quintile so that $\text{SES}_c = \text{SES} - 3$. The robustness specification replaces quintiles with standardised log equivalised income.

4.4 Additional outcomes

Two further outcomes serve the mechanism heterogeneity analysis, both measured at sweep 7 (age 17): the cohort member’s self-reported ever-smoking indicator and the Strengths and Difficulties Questionnaire (SDQ) total difficulties score (Goodman, 1997). The SDQ is a parent-reported screening instrument summing four subscales (emotional symptoms, conduct problems, hyperactivity-inattention, and peer relationship problems), with higher scores indicating greater difficulties. The trait-matched standard smoking PGI uses the GWAS of Karlsson Linnér et al. (2019); the standard SDQ PGI uses the subjective wellbeing GWAS of Okbay et al. (2016), so a higher SWB PGI corresponds to greater propensity for wellbeing and a negative coefficient on SDQ total difficulties is the expected direction. The DGE counterparts use the family-GWAS weights of Tan et al. (2026). Because both outcomes are only available at sweep 7, the mechanism heterogeneity analyses use the age 17 cross-section.

4.5 Control variables

All specifications from Column (3) onward include sex, MCS cohort indicators, and the first ten genetic principal components. The MCS genetic sample does not contain direct measures of parental health behaviours, dietary provision, or household food environments; the family channel is therefore identified from the Standard-vs-DGE PGI contrast and from parental PGI controls.

4.6 Descriptive statistics

Table 3 reports summary statistics for the estimation sample of 5,446 children. Mean BMI rises from 16.34 at age 5 to 23.41 at age 17, and within-wave dispersion widens with age. The sample skews modestly toward higher-SES households, consistent with the genotype-consent attrition pattern documented for the MCS.

5 Estimation

The empirical strategy proceeds in six steps. Step 1 builds the canonical Standard PGI $\times$ SES specification with progressively richer controls (§5.1–§5.4). Step 2 re-estimates the same specifications with the DGE PGI in place of the Standard PGI; because the two PGIs enter identical regressions, the DGE results are reported as Panel B of the same tables (§6.1, Panel B). Step 3 puts both PGIs into a single joint regression (§5.5). Step 4 tests, on that regression, the restriction that pure H1 implies between the two interactions: a cluster-robust Wald test of $\beta_{GS}^{\text{std}} = r \beta_{GS}^{\text{DGE}}$, with r the errors-in-variables benchmark (same subsection). Step 5 runs three falsification probes: developmental heterogeneity (§5.6), functional-form and imputation-free robustness (§5.7), and cross-outcome mechanism heterogeneity (§5.8). Step 6 synthesises the evidence in §6.7. Inference uses cluster-robust standard errors at the child level throughout.

5.1 Baseline associations

I first establish the unconditional Standard PGI–BMI association and the unconditional SES–BMI gradient, then combine them in a controlled baseline. This anchors the documented BMI gradients before interaction terms enter.

Table 3: Descriptive statistics

	N	Mean	SD	Min	Max
<i>Polygenic indices</i>					
Standard BMI PGI (child)	5,446	−0.03	1.07	−4.25	3.72
DGE BMI PGI (child)	5,446	0.01	1.00	−2.81	4.58
Mother BMI PGI	5,446	−0.05	1.03	−3.82	3.84
Father BMI PGI	5,446	−0.05	0.83	−3.97	4.63
<i>Socioeconomic status and demographics</i>					
SES quintile (1–5)	5,446	3.18	1.36	1	5
Female	5,446	0.51	0.50	0	1
Waves per child	5,446	4.77	0.54	1	5
<i>BMI by survey wave</i>					
Age 5	5,231	16.34	1.80	11.64	36.71
Age 7	5,139	16.61	2.26	10.11	32.17
Age 11	5,224	19.20	3.52	9.48	40.30
Age 14	5,314	21.54	4.14	12.53	46.49
Age 17	5,089	23.41	4.73	14.10	49.38

Notes: Polygenic indices, SES, and demographic variables are reported at the child level (one observation per child). SES quintiles are based on modified OECD equivalised household income at baseline (1 = most deprived, 5 = most affluent). The BMI panel reports wave-specific statistics; each child contributes up to five BMI observations.

Let BMI_{it} denote the body mass index of child i observed at wave t . Let PGI_i denote either the standard child BMI polygenic index or the DGE polygenic index. Let SES_i denote baseline socioeconomic status. All specifications include wave fixed effects δ_t , which absorb systematic age-related differences in BMI levels, and a vector of baseline controls X_i consisting of sex, cohort indicators, and the first ten genetic principal components to account for residual population stratification.

I begin by estimating two simple regressions that summarise unconditional associations:

$$BMI_{it} = \alpha + \beta_G PGI_i + u_{it}, \quad (26)$$

$$BMI_{it} = \alpha + \beta_S SES_i + v_{it}. \quad (27)$$

I then estimate a controlled baseline specification:

$$BMI_{it} = \alpha + \beta_G PGI_i + \beta_S SES_i + \delta_t + X_i' \gamma + \varepsilon_{it}. \quad (28)$$

The specification isolates the association between genetic predisposition and BMI net of socioeconomic background and demographic controls.

5.2 Socioeconomic moderation

With the baseline in place, I add the multiplicative $PGI \times SES$ interaction. The coefficient β_{GS} is the headline target of the rest of the paper: §6.1 estimates it first with the Standard PGI and then

with the DGE PGI on the same sample.

To examine whether the genetic association varies across the socioeconomic distribution, I estimate the canonical $\text{PGI} \times \text{SES}$ specification (22):

$$BMI_{it} = \alpha + \beta_G PGI_i + \beta_S SES_i + \beta_{GS}(PGI_i \times SES_i) + \delta_t + X_i' \gamma + \varepsilon_{it}. \quad (29)$$

When SES is centred at the middle quintile, β_G represents the genetic association at the midpoint of the income distribution, and β_{GS} captures how the strength of the genetic association changes as socioeconomic status increases. A negative β_{GS} implies weaker genetic associations at higher SES.

5.3 Conditioning on parental genetic background

I then add maternal and paternal PGIs as controls so that the child interaction coefficient is identified net of the parental layer of the family channel. Even with these controls, the Standard PGI itself still bundles direct, indirect, and stratification channels: by itself it cannot attribute the SES gradient to a specific channel. The Standard-vs-DGE comparison in §5.5 is what completes the attribution.

As established in the framework above, child PGIs are correlated with parental genotypes through inheritance, so the standard PGI bundles direct genetic effects with indirect family-level channels. To absorb these channels and implement the H1–H2 comparison, I extend the model to include maternal and paternal polygenic indices:

$$BMI_{it} = \alpha + \beta_G PGI_i + \beta_S SES_i + \beta_{GS}(PGI_i \times SES_i) + \beta_m PGI_i^m + \beta_p PGI_i^p + \delta_t + X_i' \gamma + \varepsilon_{it}. \quad (30)$$

I then estimate a more flexible specification that additionally includes interactions between the child polygenic index and parental polygenic indices, and between parental polygenic indices and socioeconomic status:

$$\begin{aligned} BMI_{it} = & \alpha + \beta_G PGI_i + \beta_S SES_i + \beta_{GS}(PGI_i \times SES_i) \\ & + \beta_m PGI_i^m + \beta_p PGI_i^p + \phi_{mc}(PGI_i^m \times PGI_i) + \phi_{pc}(PGI_i^p \times PGI_i) \\ & + \phi_{ms}(PGI_i^m \times SES_i) + \phi_{ps}(PGI_i^p \times SES_i) + \delta_t + X_i' \gamma + \varepsilon_{it}. \end{aligned} \quad (31)$$

The flexible interaction structure ensures that estimated child-level moderation is not mechanically driven by restrictions imposed on parental genetic effects. Equation (30) corresponds to Column (5) in the results tables; equation (31) corresponds to Column (7).

5.4 Nonlinearity in SES

Before the test of pure H1 in §5.5, I replace continuous SES with quintile indicators. This serves two purposes: it presents the PGI–SES interaction without a linearity assumption, and it produces the quintile-specific interaction coefficients that the test of pure H1 uses, quintile by quintile.

Let $\mathbf{1}\{SES_i = q\}$ denote membership of income quintile q . I estimate

$$\begin{aligned}
BMI_{it} = & \alpha + \beta_G PGI_i + \sum_q \pi_q \mathbf{1}\{SES_i = q\} + \sum_q \lambda_q (PGI_i \times \mathbf{1}\{SES_i = q\}) \\
& + \beta_m PGI_i^m + \beta_p PGI_i^p + \delta_t + X_i' \gamma + \varepsilon_{it}.
\end{aligned} \tag{32}$$

This yields group-specific genetic slopes and allows heterogeneity patterns to be presented without relying on linearity assumptions.

5.5 The H1–H2 contrast and design features

The canonical Standard PGI $\times$ SES interaction is negative; the DGE PGI $\times$ SES interaction is near zero. The two estimates come from separate Panel A and Panel B regressions. This subsection puts both into a single joint regression and asks whether their divergence is consistent with pure H1.

The contrast is implemented on a single regression that contains both polygenic indices and both interactions:

$$\begin{aligned}
BMI_{it} = & \alpha + \beta_G^{\text{Std}} \widehat{\text{PGI}}_i^{\text{Std}} + \beta_G^{\text{DGE}} \widehat{\text{PGI}}_i^{\text{DGE}} + \beta_S SES_i \\
& + \beta_{GS}^{\text{Std}} (\widehat{\text{PGI}}_i^{\text{Std}} \times SES_i) + \beta_{GS}^{\text{DGE}} (\widehat{\text{PGI}}_i^{\text{DGE}} \times SES_i) + \mathbf{Z}_i' \boldsymbol{\theta} + \delta_t + \varepsilon_{it},
\end{aligned} \tag{33}$$

where $\mathbf{Z}_i$ collects parental PGI controls, sex, cohort indicators, and the leading PCs.

The two interactions β_{GS}^{Std} and β_{GS}^{DGE} do not estimate the same parameter, even under pure H1. The DGE PGI is built from family-based GWAS estimates that condition on parental genotypes at the discovery stage, so its weights have already absorbed the indirect-effect and stratification contributions; its SES interaction identifies (asymptotically) the direct-channel part of the gradient alone. The Standard PGI interaction identifies the bundle of direct, indirect, and stratification contributions. Under pure H1 the indirect and stratification parts are zero, so both interactions reduce to the same underlying direct-channel interaction—but each scaled by how strongly its PGI loads on that channel, and the two loadings differ because the DGE PGI is built from a discovery sample an order of magnitude smaller. Under pure H1, therefore, the two interactions are not equal: they stand in a fixed ratio.

That ratio is what makes the contrast a test. I take it from the errors-in-variables framework of [Becker et al. \(2021\)](#). Under H1, both polygenic indices are noisy proxies for the latent direct-effect SNP factor f^{dir} , with attenuation factor

$$a^c = \text{Corr}(\widehat{\text{PGI}}^c, f^{\text{dir}}) = \sqrt{R_c^2 / h_{\text{dir}}^2},$$

where R_c^2 is the variance of BMI explained by PGI c in the estimation sample and h_{dir}^2 is the SNP heritability of the latent direct-effect factor ($h_{\text{dir}}^2 = 0.212$ for BMI from [Tan et al., 2026](#)). The ratio of the two PGIs' observable coefficients on any function of BMI—main effect or SES interaction—under H1 is then

$$\frac{\hat{\beta}^{\text{Std}}}{\hat{\beta}^{\text{DGE}}} = \frac{a^{\text{Std}}}{a^{\text{DGE}}} = \sqrt{\frac{R_{\text{Std}}^2}{R_{\text{DGE}}^2}}. \tag{34}$$

Pure H1 therefore implies the restriction $\beta_{GS}^{\text{Std}} = r \beta_{GS}^{\text{DGE}}$ with $r = \sqrt{R_{\text{Std}}^2 / R_{\text{DGE}}^2}$. The test is a two-sided cluster-robust Wald test of that restriction, with inference clustered at the child level as in the main regressions (30); Section 6 reports it with SES entering as quintile indicators.

If the observed $\hat{\beta}_{GS}^{\text{Std}} / \hat{\beta}_{GS}^{\text{DGE}}$ matches r , the data are consistent with pure H1. If the observed ratio is substantially larger than r , pure H1 cannot account for the divergence: the Standard PGI's interaction carries family-channel signal that the DGE PGI does not, pointing to a non-zero indirect-channel contribution (H2, or an H1+H2 mixture). A failure to reject, by contrast, does not establish pure H1—it is equally what occurs when the DGE PGI is too noisy for the contrast to have power, the case §6.3 shows obtains here. Becker et al. (2021) note that measurement-error correction tends to increase, not decrease, the magnitude of PGI-by-environment interactions, so the benchmark is conservative in the direction of understating the family-channel share.

5.6 Age heterogeneity

A family-channel reading would attribute the Standard PGI $\times$ SES gradient to household pathways. If so, the gradient should be largest where household-mediated exposure has accumulated most (late adolescence). The panel structure allows this developmental profile of the Standard gradient to be described by letting moderation vary by wave; it characterises the contrast rather than identifying its channel.

Let $\mathbf{1}\{t = s\}$ denote an indicator for wave $s \in \{5, 7, 11, 14, 17\}$. I estimate

$$\begin{aligned} \text{BMI}_{it} = & \alpha + \beta_G \text{PGI}_i + \beta_S \text{SES}_i + \beta_{GS} (\text{PGI}_i \times \text{SES}_i) \\ & + \sum_s \delta_s \mathbf{1}\{t = s\} + \sum_s \gamma_s^G (\text{PGI}_i \times \mathbf{1}\{t = s\}) + \sum_s \gamma_s^S (\text{SES}_i \times \mathbf{1}\{t = s\}) \\ & + \sum_s \gamma_s^{GS} (\text{PGI}_i \times \text{SES}_i \times \mathbf{1}\{t = s\}) + \beta_m \text{PGI}_i^m + \beta_p \text{PGI}_i^p + \mathbf{X}_i' \boldsymbol{\gamma} + \varepsilon_{it}, \end{aligned} \quad (35)$$

with one wave (the youngest) absorbed into the constant. The wave-specific deviation γ_s^{GS} measures how the PGI $\times$ SES interaction at wave s differs from its value at the reference wave; the implied wave- s slope is $\beta_{GS} + \gamma_s^{GS}$. The wave-specific structure yields PGI $\times$ SES slopes that vary across ages without imposing constancy.

5.7 Robustness checks

Three robustness checks test whether the Standard-vs-DGE contrast is stable under alternative functional forms and an imputation-free sample.

Functional form of the polygenic index. The polygenic index enters all specifications linearly. I assess this assumption by augmenting the canonical specification with a quadratic PGI term and its interaction with SES:

$$\text{BMI}_{it} = \alpha + \beta_G \text{PGI}_i + \beta_{G^2} \text{PGI}_i^2 + \beta_S \text{SES}_i + \beta_{GS} (\text{PGI}_i \times \text{SES}_i) + \beta_{G^2S} (\text{PGI}_i^2 \times \text{SES}_i) + \dots$$

A quadratic PGI $\times$ SES interaction that swamps the linear term would force a non-linear-in-PGI reading; one that does not would support the linear specification as the primary form.

Functional form of SES. The primary specification uses income-quintile rank. As a robustness check I re-estimate the canonical $\text{PGI} \times \text{SES}$ interaction with SES replaced by standardised log equivalised household income (§4.3). A negative interaction that depends on the linear-in-quintile parameterisation would be a fragile result; one that survives the change of functional form is not.

Imputed versus directly observed parental genotypes. The main sample uses Mendelian-imputed parental genotypes for children whose paternal or maternal genotype is missing (§3.2.3). As a robustness check I restrict the sample to children with both parents directly genotyped and rebuild the DGE PGI on this restricted set, so neither the analysis sample nor the DGE PGI carries any imputed parental data. If imputation generates the Standard-vs-DGE contrast through measurement error, the restricted-sample contrast should weaken or reverse.

5.8 Mechanism heterogeneity across outcomes

If the BMI Standard-vs-DGE contrast reflects a real channel difference rather than a mechanical property of regressing any polygenic score on a socially patterned outcome, the contrast should not appear for outcomes whose causal pathways differ from BMI’s household-mediated pathway. I apply the same comparison to two age-17 outcomes (§4.4): an ever-smoking indicator and the SDQ total difficulties score. A non-zero $\text{PGI} \times \text{SES}$ interaction common to both predictors for any outcome would point to a mechanical artefact; a contrast that varies with the outcome’s biological and social pathway is informative about the underlying mechanism.

6 Results

The results follow the six-step structure of §5. Step 1 confirms the negative Standard $\text{PGI} \times \text{SES}$ gradient (§6.1, Panel A). Step 2 shows the DGE counterpart is statistically indistinguishable from zero (Panel B of the same table). Step 3 puts both interactions into a single joint regression (§6.3); the quintile parametrisation of §6.2 supplies the coefficients the test uses. Step 4 tests, on that regression, the restriction that pure H1 implies between the two interactions, in the same subsection. Step 5 reports the three falsification probes: developmental (§6.4), functional-form and imputation-free robustness (§6.5), and cross-outcome mechanism heterogeneity (§6.6). Step 6 synthesises in §6.7.

6.1 Baseline gradients in BMI by genetics and socioeconomic status

Table 5 reports a stepwise sequence of pooled specifications for the Standard child BMI PGI (Panel A) and the child DGE PGI (Panel B). All models include wave fixed effects and the standard control set, and inference is based on standard errors clustered at the child level.

Panel A: Standard PGI. Column (1) establishes that a one-standard-deviation increase in the Standard BMI PGI is associated with an increase of approximately 0.9 BMI units. Column (2) confirms the SES gradient: each one-quintile increase in income rank corresponds to a reduction of 0.30 BMI units. In Column (3), both predictors enter jointly with stable coefficients, indicating that the genetic and socioeconomic gradients are largely independent.

Column (4) introduces the child PGI $\times$ SES interaction. The coefficient is negative (-0.042), implying that the PGI–BMI association is steeper at lower income rank: the implied PGI slope ranges from 1.04 in the lowest quintile to 0.87 in the highest, a bottom-to-top difference of 0.17 BMI units per standard deviation. This coefficient is not statistically significant at conventional levels, but the sign and magnitude are stable as additional controls are introduced.

Columns (5) through (7) progressively incorporate parental PGIs and their interactions. In Column (7), the most complete specification, the child PGI $\times$ SES coefficient reaches -0.070 and is statistically significant at the 5 percent level. The implied PGI slope is 1.04 in quintile 1 and 0.76 in quintile 5, a bottom-to-top difference of 0.28 BMI units per standard deviation. The father PGI $\times$ SES coefficient in this specification is positive and significant at the 5 percent level (0.096). The implied slope of BMI on the father PGI, however, changes sign across the income distribution: -0.22 at quintile 1 ($p = 0.06$), -0.03 at quintile 3, and $+0.17$ at quintile 5 ($p = 0.02$, the only individually significant point). Although the absolute slope is larger at the bottom of the distribution than at the top ($|Q1|/|Q5| \approx 1.33$), the pattern is not the same-sign, monotone signature that direct family-channel moderation would predict; the only quintile at which the slope is individually significant lies at the top of the income distribution. We therefore treat this coefficient as auxiliary and do not interpret it as evidence for H2. Across Columns (4) through (7), the negative sign of the child interaction is consistent, and its magnitude and precision increase as the specification more fully accounts for parental genetic pathways.

Panel B: DGE PGI. The DGE PGI isolates the child’s directly inherited genetic variation, purged of parental and stratification channels. In Column (3), the DGE PGI–BMI association is 0.142 BMI units per standard deviation—substantially smaller than the standard PGI coefficient, consistent with the standard PGI capturing additional family-level genetic signal. When parental PGIs are introduced in Column (5), the maternal and paternal coefficients are 0.584 and 0.428 respectively, confirming that parental genotypes independently predict child BMI above and beyond the child’s own direct genetic effects.

The key result is the interaction. Across Columns (4) through (7), the DGE PGI $\times$ SES coefficient is close to zero and not statistically significant in any specification. The implied DGE PGI slopes at the bottom and top of the income distribution are correspondingly similar, with no systematic ordering by socioeconomic status.

The contrast between Panels A and B is the central result. The Standard PGI shows a negative income gradient that grows in magnitude and precision as parental genetic pathways are accounted for; the DGE PGI shows no corresponding gradient. This divergence is the observable pattern predicted under H2 (and under an H1+H2 mixture in which the family channel dominates); it is also the pattern predicted under pure H1 with a large direct-channel weight gap between the two PGIs. §6.3 asks which of these readings the data support by testing the restriction that pure H1 implies.

6.2 Nonlinear SES heterogeneity

Before the test of pure H1 in the next subsection, this subsection replaces the linear SES interaction with a quintile-dummy specification. The quintile parametrisation produces the quintile-specific coefficients that §6.3 uses to localise the channel difference at the top of the income distribution.

The linear interaction specification constrains the PGI slope to change at a constant rate across the income distribution. Relaxing this assumption allows the data to reveal whether moderation is concentrated at the bottom of the distribution—as a resource-constraint mechanism would predict—or distributed more evenly.

Table 8 replaces the linear SES interaction with interactions between the child PGI and SES quintile indicators, with quintile 1 as the reference group. This specification yields quintile-specific PGI slopes and therefore provides an interpretation based on differences in PGI slopes across discrete points of the income distribution.

For the Standard PGI, the estimated PGI slope in quintile 1 equals 1.049 BMI units per standard deviation. The implied slopes in quintiles 2 to 5 are obtained by adding the corresponding interaction coefficients to the quintile 1 slope. Under this parameterisation, the implied slope in quintile 5 equals 0.728, and the implied difference between quintile 1 and quintile 5 slopes equals 0.321 BMI units per standard deviation. Figure 2 visualises the full quintile profile: Panel A shows the steady decline in the Standard PGI slope across quintiles, while Panel B shows a flat DGE PGI slope centred near zero at every quintile.

For the DGE PGI, the implied quintile-specific slopes do not exhibit a monotone ordering from quintile 1 to quintile 5, confirming that there is no systematic socioeconomic gradient in the DGE PGI–BMI association.

6.3 The H1–H2 contrast and the test of pure H1

§6.1 produced separate Standard and DGE estimates (Panels A and B) that diverge. This subsection puts both into the single joint regression (33) and asks whether that divergence is consistent with pure H1.

Under pure H1 the two interactions are not equal: they stand in the ratio $r = \sqrt{R_{\text{Std}}^2/R_{\text{DGE}}^2}$ of equation (34), which in the estimation sample is $r \approx 10.7$. Pure H1 is therefore the restriction $\beta_{GS}^{\text{Std}} = r \beta_{GS}^{\text{DGE}}$, and the test is a cluster-robust Wald test of that restriction, with SES entering as quintile indicators.

The restriction is not rejected at any income quintile. At Q5—where the Standard PGI’s gradient is steepest—the discrepancy is $\hat{\beta}_{GS}^{\text{Std}} - r \hat{\beta}_{GS}^{\text{DGE}} = -1.7$ with $p = 0.25$.

The test does not reject because it has no power, not because it favours pure H1. The DGE PGI explains $R_{\text{DGE}}^2 \approx 0.0004$ of the variance of BMI, so its SES interaction is estimated with wide error; once that error is scaled by $r \approx 10.7$, the H1-implied value of β_{GS}^{Std} carries a confidence interval far too wide to exclude the observed estimate. A non-rejection at this precision does not distinguish pure H1 from an H1+H2 mixture.

One caveat guards against over-reading the benchmark r . For a univariate regression of BMI on a variance-standardised PGI, $R_c^2 = \hat{\beta}_c^2/\text{Var}(\text{BMI})$, so $\sqrt{R_{\text{Std}}^2/R_{\text{DGE}}^2}$ and the main-effect ratio $\hat{\beta}_G^{\text{Std}}/\hat{\beta}_G^{\text{DGE}}$ are the same quantity computed two ways; both equal 10.7 in the estimation sample. Their agreement is an algebraic identity, not independent evidence that the benchmark is the correct yardstick for the interaction.

What the contrast establishes is therefore precise, and it falls into two parts. As a finding about childhood BMI: the gradient is real and robust in the Standard PGI, and null in the DGE PGI. As

a finding about the design: with a DGE PGI built from an 82,000-family discovery sample, a null DGE interaction does not adjudicate between H2 and H1, because it is equally consistent with no direct-channel moderation and with direct-channel moderation the DGE PGI is too underpowered to detect. The H1/H2 split is thus a well-defined open question with a clear requirement for an answer: a family GWAS large enough to raise R^2_{DGE} to the point where a test of restriction (34) has power. The robustness checks and cross-outcome comparisons that follow characterise the gradient and confirm it is a genuine trait-specific phenomenon; they do not turn on the unresolved decomposition.

6.4 Age heterogeneity in SES moderation

If the Standard PGI’s SES gradient operates through household-mediated exposure, it should concentrate where that exposure has accumulated most (late adolescence). This subsection characterises the developmental profile of the Standard gradient; it describes the contrast rather than identifying its channel, since the DGE interaction is underpowered at every age.

Table 7 allows the child PGI slope, the SES gradient, and their interaction to vary by age wave using interactions with age indicators. The reported coefficients imply age-specific PGI slopes at any SES value and SES-specific PGI slopes at any age wave, without imposing that moderation is constant across development.

For the Standard PGI, the age interactions imply an increase in the PGI slope from age 5 to ages 7, 11, 14, and 17. The SES gradient also becomes more negative at older ages. The three-way interaction at age 17 is negative and statistically significant (-0.093 , $p = 0.018$), implying that the socioeconomic difference in the Standard PGI slope is larger in late adolescence than in early childhood. At earlier ages, the corresponding three-way terms are small and not statistically significant, so the age-heterogeneity specification locates the SES-graded difference in the Standard PGI slope primarily at age 17. Figure 1 visualises these implied slopes: Panel A (Standard PGI) shows the three SES-specific lines diverging with age, while Panel B (DGE PGI) shows them bunched together near zero at every age.

For the DGE PGI, the age interactions are small and the three-way terms do not imply a systematic ordering of the DGE PGI slope by income rank at any age. The developmental Standard-PGI gradient concentrates in late adolescence—consistent with a household-accumulation reading—against a DGE interaction that is flat at every age, though the DGE flatness is equally consistent with that PGI being too underpowered to register a gradient.

6.5 Robustness checks

Three checks confirm the main contrast against alternative functional forms and an imputation-free sample.

Functional form of the polygenic index. Augmenting the canonical specification with a quadratic PGI term and its interaction with SES leaves the linear $\text{PGI} \times \text{SES}$ coefficient unchanged in sign and magnitude. For the Standard PGI, the quadratic main effect is marginally significant ($\hat{\beta}_{G^2} = 0.06$, $p = 0.02$), indicating mild convexity in the PGI–BMI gradient; the quadratic interaction coefficient $\hat{\beta}_{G^2S}$ is not statistically significant. The linear specification is the appropriate primary form.

Functional form of SES. Table 6 repeats the stepwise sequence with income-quintile rank replaced by standardised log equivalised income. The Standard PGI $\times$ income interaction remains negative but is not statistically significant at conventional levels, reflecting the loss of power when SES enters as a single linear contrast. The DGE PGI $\times$ income interaction remains close to zero. The qualitative comparison—negative for the Standard predictor, null for the DGE predictor—is consistent with the baseline results.

Imputed versus directly observed parental genotypes. The main sample uses Mendelian-imputed parental genotypes (Young et al., 2022) for children whose paternal or maternal genotype is missing. Restricting to a trio sample with both parents directly genotyped yields 2,707 unique children in Panel A and 2,697 in Panel B (Table 11); the DGE PGI is recomputed on this subsample so that neither the sample definition nor the DGE PGI relies on imputed parental data. The restriction does not weaken the central pattern; it strengthens it. The Standard PGI $\times$ SES interaction in Column (7) is -0.129 ($p = 0.023$), roughly twice the corresponding full-sample estimate of -0.070 ; the implied PGI slope is 1.10 in the lowest income quintile and 0.58 in the highest, a bottom-to-top difference of 0.52 BMI units per standard deviation. The DGE PGI $\times$ SES interaction is $+0.005$ ($p = 0.95$), an even cleaner null than in the full sample. Imputation introduces conservative attenuation rather than spurious moderation: the Standard PGI $\times$ SES coefficient nearly doubles when imputation is removed, consistent with classical measurement error in the imputed parental genotypes biasing the full-sample estimate toward zero.

The trio subsample is not a random subset of the full estimation sample. Table A2 in the appendix compares the 2,707 trio children with the 2,739 children whose inclusion required Mendelian imputation of at least one parental genotype. Trio children come from higher-income households (mean quintile 3.45 versus 2.91, $p < 0.001$) with lower Standard BMI PGIs (-0.085 versus $+0.033$, $p < 0.001$) and lower mean BMI (19.27 versus 19.65, $p < 0.001$)—the well-documented selection of more stable, higher-income families into both-parent genotype consent. The DGE BMI PGI, however, is balanced between the two groups (mean -0.008 versus $+0.018$, $p = 0.33$); selection into the trio subsample is on the socioeconomic dimension, not on the direct-effect genetic dimension. Because the trio sample systematically excludes the most disadvantaged families (the quintile-1 share drops from 20.7% to 8.8%), it compresses the range where a family-channel reading would predict the strongest moderation; the strengthening of the Standard interaction under an SES-compressed sample makes it unlikely that sample selection generates the Standard-vs-DGE contrast.

6.6 Mechanism heterogeneity across outcomes

§6.3 established that the BMI gradient is carried by the Standard PGI but not the DGE PGI, and what separating its two channels would require. This subsection asks whether the same Standard-vs-DGE divergence appears for other outcomes—in which case it would be a mechanical artefact of regressing any polygenic score on any socially patterned outcome—or whether it is trait-specific. If moderation were a mechanical feature of regressing any polygenic predictor on a socially patterned outcome, the same signature would replicate for every trait; if instead the mechanism varies across traits in theoretically interpretable ways, the cross-outcome comparison itself becomes informative. I apply the same Standard-vs-DGE design to smoking behaviour and child behavioural and emotional difficulties (SDQ); the PGIs and outcome measures are described in §4.4.

The three outcomes together yield a 3×2 pattern of Standard-vs-DGE contrasts that is both

internally consistent and substantively meaningful:

Smoking behaviour. Table 9 reports the stepwise models for ever smoking. Panel A shows that the Standard smoking PGI is positively associated with ever smoking with a strong SES gradient, but the $\text{PGI} \times \text{SES}_c$ interaction is close to zero across all specifications. Panel B finds the same null interaction for the DGE smoking PGI. For smoking, neither predictor exhibits SES moderation—enough to rule out the alternative that significant $\text{PGI} \times \text{SES}$ interactions are a mechanical feature of regressing any polygenic score on any socially patterned outcome.

Child behavioural and mental health difficulties. Table 10 reports results for SDQ total difficulties using the SWB polygenic predictors described above. Panel A shows that the Standard SWB PGI is negatively associated with SDQ difficulties and there is a strong SES gradient, but the $\text{PGI} \times \text{SES}$ interaction is small and not statistically significant—the same null as the standard smoking PGI. Panel B shows a directionally different pattern: the DGE SWB $\text{PGI} \times \text{SES}$ interaction is negative in all specifications (point estimates between -0.059 and -0.071), but not statistically significant at conventional levels. The sign is consistent with children whose direct genetic endowment for wellbeing is higher showing a more protective association with behavioural difficulties at lower income rank, but the smaller cross-sectional sample at age 17 does not provide sufficient precision to confirm this pattern. The contrast with the Standard PGI is nevertheless noteworthy: moderation point estimates are concentrated in the DGE predictor rather than the Standard predictor, the reverse of the BMI pattern.

Cross-outcome pattern. Table 4 collects the three outcomes side by side. The smoking null rules out the mechanical-artefact explanation for the BMI result. The SDQ point estimates concentrate the moderation in the DGE predictor—the reverse of the BMI pattern—which is consistent with direct-effect moderation operating where the biology runs through the child’s own neurodevelopment rather than through the household; the cross-sectional sample at age 17 does not, however, permit a definitive H1 statement for SDQ. The BMI result, in which the Standard-vs-DGE contrast holds across every specification and functional form, remains the strongest evidence in the table.

Table 4: Cross-outcome pattern of Standard-vs-DGE moderation contrasts

Outcome	Standard $\text{PGI} \times \text{SES}$	DGE $\text{PGI} \times \text{SES}$	Reading of the contrast
BMI	Negative (sig. in Col. 7)	Near zero	Robust contrast; not decomposed
Smoking	Near zero	Near zero	No moderation
SDQ difficulties	Near zero	Negative (n.s.)	Suggestive of H1

Notes: Each outcome uses a trait-matched standard PGI and DGE PGI estimated from the same stepwise specifications. Smoking and SDQ analyses use the age 17 cross-section (single observation per child); BMI uses the pooled age 5–17 panel. “Sig.” denotes statistically significant; “n.s.” denotes not statistically significant at conventional levels. The SDQ point estimates are directionally consistent with H1 (direct-effect moderation) but imprecisely estimated in the cross-sectional sample.

6.7 Mechanism assessment

The results form a consistent descriptive pattern. The Standard-vs-DGE divergence holds across linear-rank, log-income, and quintile-dummy parameterisations of SES, and it strengthens, rather

than weakens, in the trio sample with all imputed parental genotypes removed. What the pattern does not deliver is a decomposition: the test of pure H1 in §6.3 does **not** rule out the alternative that the Standard-DGE gap at Q5 reflects only the precision-penalty attenuation of a pure-H1 process—the DGE PGI is too imprecise for the pure-H1 restriction to be rejected. The cross-outcome comparison adds external evidence: when the Standard-vs-DGE design is applied to smoking, neither predictor moderates by SES, which removes the worry that the BMI result is a mechanical property of interacting any polygenic score with any socially patterned outcome. The SDQ point estimates concentrate the moderation in the DGE rather than the Standard predictor, which is consistent with direct-effect moderation operating where the biology runs through the child’s own neurodevelopment rather than through the household environment.

Two qualifications dominate the reading. The DGE PGI is a noisy construct: it explains about 0.04% of the variance of BMI, and at that precision its SES interaction cannot discriminate the absence of direct-effect moderation from direct-effect moderation it is simply too weak to detect. The constant-coefficient outcome model also commits the analysis to a population-mean reading of the direct effect; the SES-specific reading a strong biological H1 would target lies outside what the family-GWAS weights estimate. The position the analysis supports for childhood BMI in the MCS is therefore not a decomposition into channels but the narrower, descriptive claim that the SES gradient is carried by the standard PGI and not the DGE PGI, with the household-versus-genome question left open for a larger family GWAS to settle.

Table 5: Child PGIs, socioeconomic status, and BMI (Clustered)

	Dependent variable: BMI_{it}						
	(1) PGI	(2) SES	(3) + Controls	(4) + PGI×SES	(5) + Parents	(6) + Parent×Child	(7) + Parent×SES
Panel A: Standard PGI							
Child PGI	0.921*** (0.036)		0.947*** (0.038)	0.956*** (0.039)	0.883*** (0.049)	0.889*** (0.050)	0.900*** (0.051)
SES _c		-0.303*** (0.030)	-0.189*** (0.029)	-0.189*** (0.029)	-0.181*** (0.029)	-0.180*** (0.029)	-0.176*** (0.029)
Mother PGI					0.148*** (0.045)	0.151*** (0.046)	0.145*** (0.047)
Father PGI					0.013 (0.048)	0.011 (0.049)	-0.027 (0.055)
Child PGI × SES _c				-0.042 (0.028)	-0.040 (0.027)	-0.028 (0.028)	-0.070** (0.036)
Mother PGI × Child PGI						0.070** (0.034)	0.070** (0.035)
Father PGI × Child PGI						0.016 (0.037)	0.028 (0.037)
Mother PGI × SES _c							0.030 (0.035)
Father PGI × SES _c							0.096** (0.040)
Observations	5,446	5,446	5,446	5,446	5,446	5,446	5,446
R^2	0.049	0.009	0.442	0.442	0.443	0.443	0.444
Panel B: DGE PGI							
Child DGE PGI	0.092** (0.039)		0.142** (0.066)	0.141** (0.068)	0.096 (0.066)	0.096 (0.067)	0.095 (0.067)
SES _c		-0.303*** (0.030)	-0.289*** (0.030)	-0.289*** (0.030)	-0.210*** (0.030)	-0.211*** (0.030)	-0.210*** (0.030)
Mother PGI					0.584*** (0.040)	0.584*** (0.040)	0.590*** (0.041)
Father PGI					0.428*** (0.044)	0.429*** (0.044)	0.406*** (0.049)
DGE PGI × SES _c				0.004 (0.030)	-0.005 (0.030)	-0.009 (0.031)	-0.008 (0.031)
Mother PGI × DGE PGI						-0.046 (0.041)	-0.048 (0.041)
Father PGI × DGE PGI						0.043 (0.042)	0.046 (0.042)
Mother PGI × SES _c							-0.028 (0.030)
Father PGI × SES _c							0.049 (0.036)
Observations	5,446	5,446	5,446	5,446	5,446	5,446	5,446
R^2	0.000	0.009	0.397	0.397	0.419	0.419	0.419

Notes: Columns (1) and (2) are bivariate regressions. Columns (3) to (7) additionally include age fixed effects, sex, cohort indicators, and the first ten genetic principal components. Socioeconomic status is the OECD equivalised income quintile score centred at 3, so $SES_c = 0$ corresponds to quintile 3. Standard errors in parentheses are clustered at the child level. Significance: * 10%, ** 5%, *** 1%.

Table 6: Child PGIs, equivalised household income, and BMI (Clustered)

	Dependent variable: BMI_{it}						
	(1) PGI	(2) Income	(3) + Controls	(4) + PGI×Income	(5) + Parents	(6) + Parent×Child	(7) + Parent×Income
Panel A: Standard PGI							
Child PGI	0.923*** (0.036)		0.955*** (0.038)	0.964*** (0.040)	0.890*** (0.050)	0.896*** (0.050)	0.906*** (0.051)
SES_use		-0.391*** (0.043)	-0.233*** (0.041)	-0.232*** (0.041)	-0.222*** (0.041)	-0.223*** (0.041)	-0.216*** (0.041)
Mother PGI					0.150*** (0.045)	0.153*** (0.046)	0.147*** (0.047)
Father PGI					0.009 (0.048)	0.007 (0.049)	-0.031 (0.055)
Child PGI × SES_use				-0.046 (0.038)	-0.043 (0.038)	-0.027 (0.039)	-0.085* (0.051)
Mother PGI × Child PGI						0.072** (0.035)	0.072** (0.035)
Father PGI × Child PGI						0.019 (0.037)	0.030 (0.037)
Mother PGI × SES_use							0.040 (0.049)
Father PGI × SES_use							0.131** (0.057)
Observations	5,406	5,406	5,406	5,406	5,406	5,406	5,406
Panel B: DGE PGI							
Child DGE PGI	0.093** (0.039)		0.154** (0.067)	0.145** (0.069)	0.098 (0.067)	0.099 (0.067)	0.097 (0.067)
SES_use		-0.391*** (0.043)	-0.371*** (0.043)	-0.371*** (0.043)	-0.265*** (0.043)	-0.266*** (0.043)	-0.263*** (0.043)
Mother PGI					0.592*** (0.040)	0.592*** (0.040)	0.597*** (0.042)
Father PGI					0.429*** (0.044)	0.430*** (0.044)	0.404*** (0.050)
DGE PGI × SES_use				0.044 (0.045)	0.032 (0.044)	0.028 (0.045)	0.029 (0.045)
Mother PGI × DGE PGI						-0.040 (0.042)	-0.041 (0.042)
Father PGI × DGE PGI						0.046 (0.042)	0.049 (0.043)
Mother PGI × SES_use							-0.031 (0.042)
Father PGI × SES_use							0.077 (0.050)
Observations	5,406	5,406	5,406	5,406	5,406	5,406	5,406

Notes: Columns (1) and (2) are bivariate regressions. Columns (3) to (7) additionally include age fixed effects, sex, cohort indicators, and the first ten genetic principal components. Socioeconomic status is measured by modified OECD equivalised household income and enters as a standardised log variable. Standard errors in parentheses are clustered at the child level. Significance: * 10%, ** 5%, *** 1%.

Table 7: Age heterogeneity in SES moderation

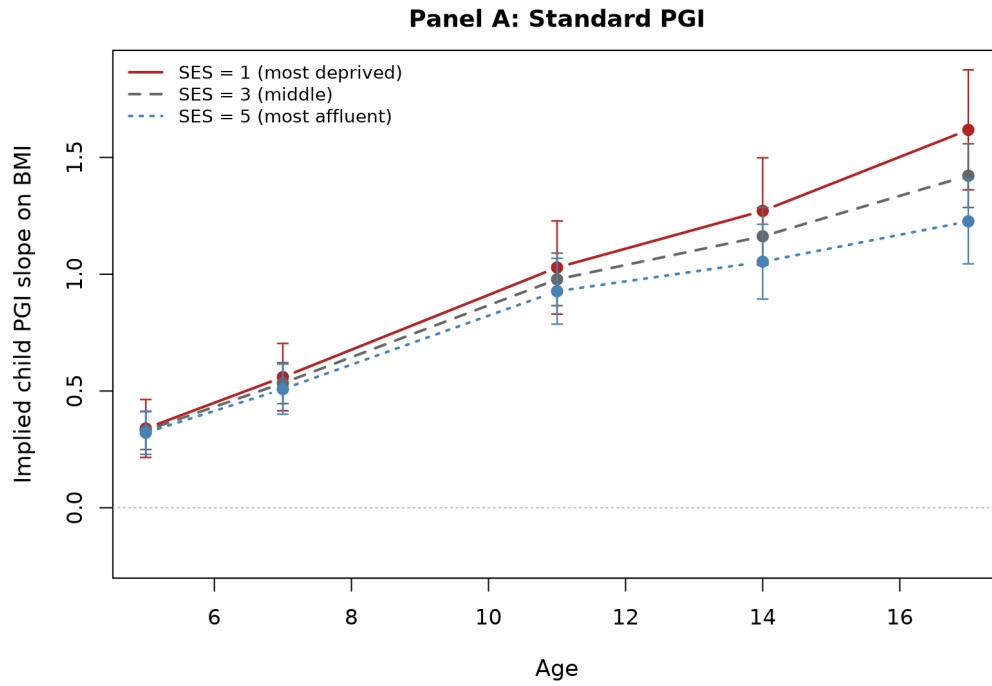
	Standard PGI	DGE PGI
Child PGI (Age 5)	0.330*** (0.041)	0.070 (0.058)
SES (Age 5)	-0.025 (0.019)	0.020 (0.020)
Child PGI $\times$ Age 7	0.203*** (0.023)	0.015 (0.023)
Child PGI $\times$ Age 11	0.648*** (0.039)	0.039 (0.041)
Child PGI $\times$ Age 14	0.832*** (0.047)	0.054 (0.050)
Child PGI $\times$ Age 17	1.092*** (0.056)	0.023 (0.062)
SES $\times$ Age 7	-0.019 (0.016)	-0.049*** (0.017)
SES $\times$ Age 11	-0.171*** (0.028)	-0.259*** (0.029)
SES $\times$ Age 14	-0.302*** (0.036)	-0.414*** (0.037)
SES $\times$ Age 17	-0.285*** (0.042)	-0.425*** (0.044)
Child PGI $\times$ SES $\times$ Age 7	-0.008 (0.015)	-0.001 (0.017)
Child PGI $\times$ SES $\times$ Age 11	-0.021 (0.026)	0.004 (0.029)
Child PGI $\times$ SES $\times$ Age 14	-0.050 (0.034)	-0.011 (0.036)
Child PGI $\times$ SES $\times$ Age 17	-0.093** (0.040)	0.008 (0.044)
Mother PGI	0.147*** (0.045)	0.584*** (0.040)
Father PGI	0.011 (0.048)	0.427*** (0.044)
Observations	5,446	5,446

Notes: All models include age fixed effects, cohort indicators, sex, and the first ten genetic principal components. Standard errors in parentheses are clustered at the child level. Significance: * 10%, ** 5%, *** 1%.

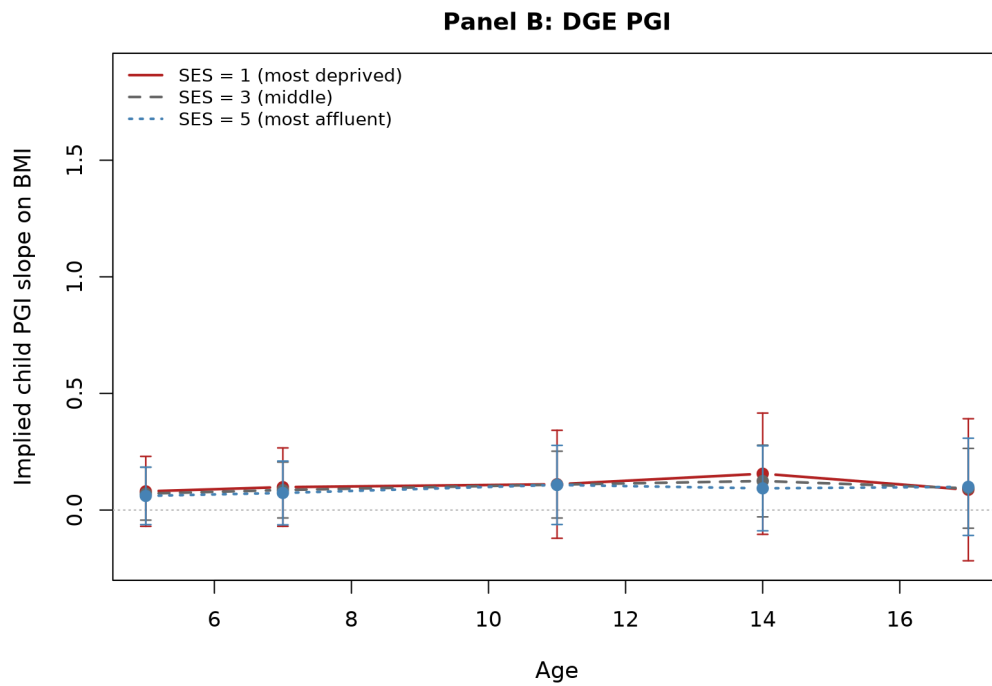
Table 8: Nonlinear SES heterogeneity

	Standard PGI	DGE PGI
Child PGI (Quintile 1 slope)	1.049*** (0.116)	-0.047 (0.134)
Child PGI $\times$ Quintile 2	-0.250* (0.143)	0.246 (0.161)
Child PGI $\times$ Quintile 3	-0.063 (0.136)	0.259* (0.144)
Child PGI $\times$ Quintile 4	-0.178 (0.131)	0.038 (0.137)
Child PGI $\times$ Quintile 5	-0.322** (0.131)	0.131 (0.141)
SES quintile 2	0.013 (0.148)	-0.142 (0.156)
SES quintile 3	-0.024 (0.139)	-0.185 (0.147)
SES quintile 4	-0.264* (0.136)	-0.409*** (0.143)
SES quintile 5	-0.711*** (0.138)	-0.881*** (0.142)
Mother PGI	0.146*** (0.045)	0.585*** (0.040)
Father PGI	0.013 (0.048)	0.427*** (0.043)
Observations	5,446	5,446

Notes: SES quintile 1 is the reference group. All models include age fixed effects, cohort indicators, sex, and the first ten genetic principal components. Standard errors in parentheses are clustered at the child level. Significance: * 10%, ** 5%, *** 1%.



(a) Standard PGI



(b) DGE PGI

Figure 1: Age heterogeneity in SES moderation³⁵: implied child PGI slopes by age and SES

Notes: The figure reports implied marginal effects of the child PGI on BMI evaluated at SES values 1, 3, and 5 for each age. Implied slopes are computed from a specification that includes wave fixed effects, the full control set, parental PGIs, and interactions that allow the $PGI \times SES$ gradient to vary by wave. Vertical bars indicate 95% confidence intervals based on standard errors clustered at the child level.

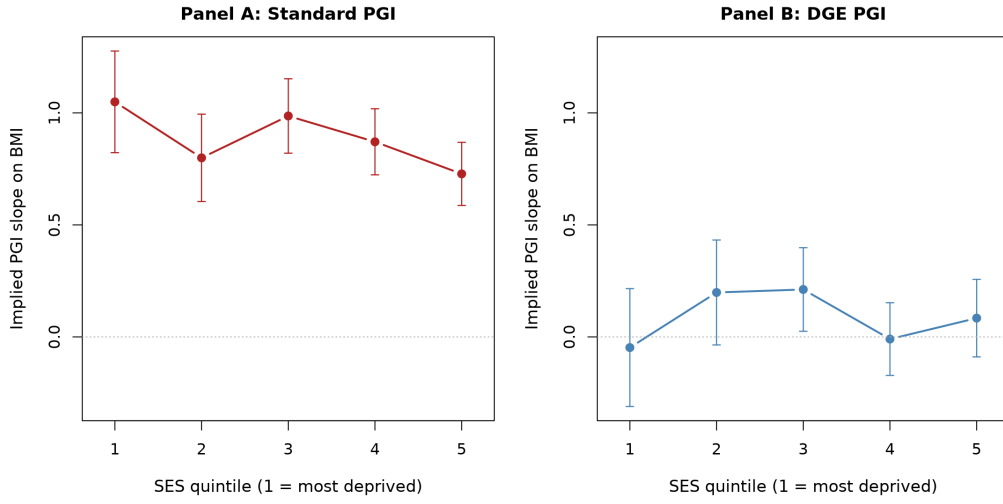


Figure 2: Nonlinear SES heterogeneity: implied child PGI slopes by SES quintile

Notes: The figure reports implied within quintile slopes of BMI on the child PGI from specifications that interact the child PGI with SES quintile indicators. Quintile 1 is the reference group. The implied slope in quintile q equals the PGI coefficient plus the interaction term for quintile q . Vertical bars indicate 95% confidence intervals based on standard errors clustered at the child level. Regressions include wave fixed effects, the full control set, and parental PGIs, and are estimated on the parents available sample for each PGI specification.

Table 9: Mechanism heterogeneity: smoking behaviour on smoking PGIs

	Dependent variable: <i>EverSmoke_i</i>						
	(1) PGI	(2) SES _c	(3) + Controls	(4) + PGI×SES _c	(5) + Parents	(6) + Parent×Child	(7) + Parent×SES _c
Panel A: Standard PGI							
Child smoking PGI	0.041*** (0.005)		0.037*** (0.005)	0.038*** (0.005)	0.028*** (0.006)	0.028*** (0.006)	0.028*** (0.006)
SES _c		-0.034*** (0.004)	-0.033*** (0.004)	-0.033*** (0.004)	-0.032*** (0.004)	-0.032*** (0.004)	-0.032*** (0.004)
Mother PGI					0.013** (0.005)	0.013** (0.005)	0.013** (0.006)
Father PGI					0.007 (0.005)	0.007 (0.005)	0.006 (0.006)
Child PGI × SES _c				-0.001 (0.004)	-0.001 (0.004)	-0.000 (0.004)	-0.002 (0.005)
Mother PGI × Child PGI						0.007* (0.004)	0.007* (0.004)
Father PGI × Child PGI						-0.002 (0.004)	-0.002 (0.004)
Mother PGI × SES _c							0.002 (0.004)
Father PGI × SES _c							0.002 (0.004)
Observations	5,367	5,367	5,367	5,367	5,367	5,367	5,367
R ²	0.015	0.018	0.041	0.041	0.042	0.042	0.043
Panel B: DGE PGI							
Child smoking DGE PGI	0.025*** (0.005)		0.024*** (0.005)	0.025*** (0.006)	0.019*** (0.006)	0.018*** (0.006)	0.018*** (0.006)
SES _c		-0.016*** (0.005)	-0.019*** (0.005)	-0.019*** (0.005)	-0.016*** (0.005)	-0.016*** (0.005)	-0.016*** (0.005)
Mother PGI					0.023*** (0.006)	0.023*** (0.006)	0.026*** (0.007)
Father PGI					0.018*** (0.006)	0.018*** (0.006)	0.015** (0.007)
Child PGI × SES _c				-0.001 (0.004)	-0.001 (0.004)	0.001 (0.004)	0.001 (0.004)
Mother PGI × Child PGI						0.001 (0.005)	-0.000 (0.006)
Father PGI × Child PGI						0.012** (0.006)	0.012** (0.006)
Mother PGI × SES _c							-0.006 (0.005)
Father PGI × SES _c							0.005 (0.005)
Observations	2,661	2,661	2,661	2,661	2,661	2,661	2,661
R ²	0.007	0.005	0.022	0.022	0.031	0.032	0.033

Notes: Columns (1) and (2) are bivariate regressions. Columns (3) to (7) additionally include sex, cohort indicators, and the first ten genetic principal components (PC1 to PC10). SES is the OECD equivalised household income quintile score (1 to 5) centred at the middle quintile (so $SES_c = SES - 3$). Child, mother, and father PGIs are standardised (z-scored) once within the common sample used for each panel, and the standardised versions are used throughout. Standard errors in parentheses are heteroskedasticity-robust (HC1) standard errors. Significance: * 10%, ** 5%, *** 1%.

Table 10: Mechanism heterogeneity: SDQ total difficulties on subjective wellbeing PGIs

	Dependent variable: SDQ_i (total difficulties)						
	(1) PGI	(2) SES_c	(3) + Controls	(4) + PGI $\times$ SES_c	(5) + Parents	(6) + Parent $\times$ Child	(7) + Parent $\times$ SES_c
Panel A: Standard PGI							
Child SWB PGI	-0.449*** (0.087)		-0.434*** (0.088)	-0.422*** (0.092)	-0.164 (0.119)	-0.167 (0.119)	-0.170 (0.119)
SES_c		-0.920*** (0.066)	-0.938*** (0.067)	-0.939*** (0.067)	-0.927*** (0.067)	-0.914*** (0.067)	-0.915*** (0.067)
Mother PGI					-0.375*** (0.104)	-0.373*** (0.104)	-0.391*** (0.107)
Father PGI					-0.178* (0.092)	-0.176* (0.091)	-0.123 (0.102)
Child PGI $\times$ SES_c				-0.045 (0.066)	-0.034 (0.066)	-0.021 (0.066)	-0.011 (0.088)
Mother PGI $\times$ Child PGI						-0.058 (0.071)	-0.066 (0.071)
Father PGI $\times$ Child PGI						-0.209*** (0.061)	-0.203*** (0.061)
Mother PGI $\times$ SES_c							0.063 (0.078)
Father PGI $\times$ SES_c							-0.107 (0.074)
Observations	4,141	4,141	4,141	4,141	4,141	4,141	4,141
R^2	0.006	0.048	0.058	0.058	0.061	0.063	0.064
Panel B: DGE PGI							
Child SWB DGE PGI	-0.000 (0.108)		0.010 (0.121)	0.052 (0.141)	0.100 (0.140)	0.094 (0.140)	0.095 (0.140)
SES_c		-0.731*** (0.093)	-0.730*** (0.100)	-0.727*** (0.100)	-0.720*** (0.099)	-0.718*** (0.099)	-0.724*** (0.099)
Mother PGI					-0.470*** (0.107)	-0.476*** (0.106)	-0.473*** (0.121)
Father PGI					-0.295*** (0.108)	-0.305*** (0.108)	-0.223* (0.121)
Child PGI $\times$ SES_c				-0.071 (0.091)	-0.069 (0.090)	-0.064 (0.091)	-0.059 (0.091)
Mother PGI $\times$ Child PGI						-0.146 (0.100)	-0.153 (0.100)
Father PGI $\times$ Child PGI						-0.133 (0.105)	-0.140 (0.105)
Mother PGI $\times$ SES_c							-0.013 (0.090)
Father PGI $\times$ SES_c							-0.162* (0.088)
Observations	2,294	2,294	2,294	2,294	2,294	2,294	2,294
R^2	0.000	0.030	0.036	0.036	0.046	0.048	0.049

Notes: Columns (1) and (2) are bivariate regressions. Columns (3) to (7) additionally include sex, cohort indicators, and the first ten genetic principal components (PC1 to PC10). SES is the OECD equivalised household income quintile score (1 to 5) centred at the middle quintile (so $SES_c = SES - 3$). Child, mother, and father PGIs are standardised (z-scored) once within the common sample used for each panel, and the standardised versions are used throughout. Standard errors in parentheses are conventional OLS standard errors. Significance: * 10%, ** 5%, *** 1%.

Table 11: Child PGIs, socioeconomic status, and BMI (non-imputed sample)

	Dependent variable: BMI_{it}						
	(1) PGI	(2) SES	(3) + Controls	(4) + PGI×SES	(5) + Parents	(6) + Parent×Child	(7) + Parent×SES
Panel A: Standard PGI							
Child PGI (z)	0.843*** (0.050)		0.883*** (0.050)	0.903*** (0.058)	0.791*** (0.075)	0.791*** (0.075)	0.843*** (0.080)
SES_c		-0.259*** (0.043)	-0.173*** (0.043)	-0.174*** (0.043)	-0.164*** (0.043)	-0.164*** (0.043)	-0.159*** (0.043)
Mother PGI (z)					0.191*** (0.061)	0.199*** (0.064)	0.166** (0.069)
Father PGI (z)					0.033 (0.057)	0.035 (0.058)	-0.023 (0.067)
Child PGI × SES_c				-0.037 (0.041)	-0.036 (0.040)	-0.024 (0.041)	-0.129** (0.057)
Mother PGI × Child PGI						0.056 (0.049)	0.055 (0.049)
Father PGI × Child PGI						0.021 (0.042)	0.030 (0.041)
Mother PGI × SES_c							0.072 (0.050)
Father PGI × SES_c							0.125*** (0.048)
Observations	2,707	2,707	2,707	2,707	2,707	2,707	2,707
R^2	0.043	0.006	0.445	0.445	0.446	0.447	0.448
Panel B: DGE PGI							
Child PGI (z)	0.130** (0.055)		0.207** (0.081)	0.200** (0.089)	0.225** (0.103)	0.226** (0.103)	0.230** (0.108)
SES_c		-0.257*** (0.043)	-0.257*** (0.045)	-0.257*** (0.045)	-0.259*** (0.045)	-0.259*** (0.045)	-0.259*** (0.045)
Mother PGI (z)					-0.061 (0.062)	-0.057 (0.063)	-0.053 (0.074)
Father PGI (z)					0.001 (0.064)	-0.002 (0.065)	-0.017 (0.075)
Child PGI × SES_c				0.014 (0.047)	0.015 (0.047)	0.015 (0.047)	0.005 (0.069)
Mother PGI × Child PGI						-0.008 (0.044)	-0.007 (0.044)
Father PGI × Child PGI						0.032 (0.048)	0.032 (0.048)
Mother PGI × SES_c							-0.006 (0.053)
Father PGI × SES_c							0.027 (0.054)
Observations	2,697	2,697	2,697	2,697	2,697	2,697	2,697
R^2	0.001	0.006	0.404	0.404	0.404	0.404	0.404

Notes: Non-imputed trio sample. Children restricted to those with both parents directly genotyped (no snipar imputation of parental genotypes), yielding 2,707 unique children in Panel A (Standard PGI) and 2,697 in Panel B (DGE PGI), contributing 13,057 and 13,007 child-wave observations respectively. Panel B uses child, mother, and father DGE BMI polygenic indices recomputed on the real-trio subsample without any Mendelian imputation, with per-SNP mean-centring performed within this subsample. Columns (1) and (2) are bivariate regressions. Columns (3) to (7) additionally include wave-age fixed effects (ages 5, 7, 11, 14, 17), sex, cohort indicators, and the first ten genetic principal components (PC1 to PC10). SES_c is the OECD equivalised household income quintile score (1 to 5) centred at the middle quintile ($SES_c = SES - 3$). Standard errors in parentheses are clustered at the child level (HC1). Significance: * 10%, ** 5%, *** 1%.

7 Discussion and Conclusion

This paper has established two things about the $\text{PGI} \times \text{SES}$ gradient for childhood BMI. The gradient is carried by the standard population-GWAS index but not by the family-GWAS one; and the widely cited reading of it—that disadvantage amplifies children’s own genetic risk—is, on the evidence, a hypothesis rather than a finding. The remainder of the paper turns from estimation to interpretation. This section reads the existing $\text{PGI} \times \text{SES}$ literature in light of those two results, sets out the precise condition under which the household-versus-genome question can be answered, and discusses what the Standard-versus-DGE design delivers now and what it will deliver as family-GWAS discovery samples grow.

7.1 Reinterpreting the $\text{PGI} \times \text{SES}$ literature

A large body of work reports that BMI polygenic indices associate more strongly with body weight among socioeconomically disadvantaged individuals, and interprets this pattern as evidence that adverse environments amplify the expression of genetic predispositions toward obesity (Hüls et al., 2021; Bann et al., 2022). The results of this paper suggest that interpretation is not as secure as the literature has treated it. The Standard-vs-DGE comparison shows that the observed socioeconomic gradient is carried by the standard PGI but not by the DGE PGI. The pure-H1 reading the literature has typically adopted is therefore not established by the data; but neither is it refuted, because the DGE PGI cannot resolve the channels. The amplification interpretation should be held as a hypothesis, not a finding, until a family GWAS large enough to decompose the gradient is available.

This finding speaks directly to the ambiguity identified in the existing literature. Observational cohort studies have reported mixed results: some find that BMI PGIs associate more strongly with weight under disadvantage (Hüls et al., 2021), while others find additive patterns with near-zero interaction terms (de Roo et al., 2023). My results suggest that these discrepancies may partly reflect differences in the degree to which standard PGIs capture family-level confounding across study designs and cohorts. van den Berg et al. (2025) explicitly noted that their gene–environment estimates “may be capturing the additional exposure [to confectionery] driven by the parental genetic ‘predisposition’ [for sugar]”; Uchikoshi and Conley (2021) called for future work to separate direct genetic mechanisms from genetic nurture in PGI interaction estimates. The present paper implements that separation. With the genetic predictor restricted to directly inherited effects, the $\text{PGI} \times \text{SES}$ gradient for BMI is statistically indistinguishable from zero, so the concern that the standard-PGI gradient reflects parental rather than child genetics cannot be dismissed—though, as §6.3 shows, the current DGE PGI is too imprecise to confirm that it is correct.

Prior studies that relied on standard population-based PGIs could not distinguish between direct-effect and family-channel moderation because their genetic predictor bundles direct, indirect, and confounding components into a single composite (Benjamin et al., 2024; Biroli et al., 2026). Recent quantification of indirect genetic effects supports this concern. Howe et al. (2022) find substantial within-sibship shrinkage relative to population GWAS estimates for socially-mediated traits, including 47% for educational attainment, 50% for depressive symptoms, 52% for age at first birth, and 67% for number of children. van Alten et al. (2025) find that, for SES outcomes in Dutch tax records, “about half of next-generation genetic effects reflect direct genetic inheritance”, with the remaining half operating through genetic nurture and assortative mating. This makes clear why the description of the BMI gradient as “genetic sensitivity to disadvantage” cannot be taken

as established: the gradient is equally consistent with socioeconomic variation in the family and household environments through which parental genetic background operates.

A potential objection is that BMI shows relatively limited shrinkage in within-sibship GWAS estimates compared to education or cognitive ability (Howe et al., 2022), suggesting that standard and DGE BMI PGIs should be similar. Yet even a modest indirect genetic effect component can generate differential gene–environment interaction patterns if that component is specifically correlated with SES-related household environments. The indirect channel need not be large in the main effect to be consequential in the interaction: family dietary environments and parental health behaviours are precisely the pathways that vary most across the socioeconomic distribution.

This reinterpretation does not diminish the importance of the $\text{PGI} \times \text{SES}$ gradient—it reopens the question of what the gradient means. The gradient is real: children from disadvantaged families with high-BMI-predisposed parents do gain more weight. What is no longer settled is whether the operative mechanism is what parents do (dietary provision, health behaviours, household organisation) or how children’s own alleles express under resource constraint. That distinction matters for both scientific interpretation and intervention design, which is why leaving it open—rather than assuming it—is the substantive point.

The present design also clarifies how it relates to the quasi-experimental literature. Studies that exploit compulsory schooling reforms (Barcellos et al., 2018), drinking-age discontinuities (Fletcher and Lu, 2021), prenatal exposures (van den Berg et al., 2025; Dias Pereira et al., 2025)—the latter instrumenting maternal smoking with a *CHRNA3* genetic variant rather than a policy shock—or within-family investment variation (Muslimova et al., 2024) estimate how the causal effect of an external shock varies across the polygenic distribution—genotype-moderated treatment effects. These designs answer a different identification question from $\text{PGI} \times \text{SES}$: the SES gradient in the PGI –BMI slope is informative about how the genetic-prediction signal varies with the social environment, not about how a specific environmental intervention’s effect varies by genotype. The Standard-vs-DGE comparison localises which part of the PGI signal carries the SES interaction. Their estimates, even when labelled as gene–environment interactions, recover an object that is distinct from the SES-gradient estimand and need not contradict the present result for BMI; the two literatures are best read as complementary identifications of different quantities.

7.2 Why the cross-trait patterns might differ

The three traits show different Standard-vs-DGE patterns: a robust contrast for BMI, a qualitatively different and underpowered pattern for SDQ, and a null for smoking. The design does not identify the mechanism behind any of them. It is nonetheless worth setting out, as hypotheses for future work rather than as findings, why the patterns might differ in how genetic predispositions translate into outcomes across the socioeconomic distribution.

For BMI, weight is shaped heavily by the household food environment, which is under parental control and directly constrained by household income (Autret and Bekelman, 2024). Parents who carry BMI-increasing alleles plausibly generate home dietary environments that raise child weight independently of which alleles the child inherited (Kong et al., 2018), an extension to food environments of the diet–BMI genetic architecture documented in Meddens et al. (2021). Under disadvantage, these parental channels would be less buffered by alternative food options and health resources—a route that would produce a family-channel reading of the gradient. Breinholt and

Conley (2023) provide direct evidence for a related mechanism: parental cognitively stimulating investments respond to child genotype, and this response is socioeconomically stratified—middle-SES parents reinforce genetic advantages while college-educated parents maintain high investment regardless of child genotype.

For child behavioural and emotional difficulties (SDQ), the relevant genetic pathways—cognitive regulation, emotional reactivity, stress response—operate more directly through the child’s own neurodevelopmental endowment. Marees et al. (2021) show that genetic correlates of socioeconomic status pervasively influence the shared heritability architecture of mental health traits, suggesting that SES-related genetic variation is deeply embedded in mental health phenotypes. The point estimates for the DGE SWB $\text{PGI} \times \text{SES}$ interaction are consistent with this interpretation: they are negative across specifications, suggesting that household disadvantage may amplify direct genetic differences in behavioural difficulties. However, these estimates do not reach statistical significance, so the evidence for H1 in SDQ remains suggestive rather than conclusive. A larger sample or a longitudinal measure of behavioural difficulties would be needed to confirm whether direct-effect moderation operates for mental health outcomes.

For smoking, neither the standard nor the DGE predictor exhibits meaningful SES moderation. This null result is informative precisely because it rules out a trivial explanation: it confirms that socioeconomic moderation is not a mechanical property of interacting any PGI with any socially patterned outcome. The absence of moderation for smoking contrasts with the clear pattern for BMI and the directional evidence for SDQ, and may reflect the fact that adolescent smoking initiation operates through pathways—peer influence, experimentation, access—that do not scale with household income in the same gradient-like manner as diet quality or parental health investment. This trait-specific pattern has a broader implication: gene–environment interaction is not a monolithic phenomenon. Whether socioeconomic disadvantage amplifies genetic predispositions depends on which biological and social pathways connect genotype to phenotype, and these pathways differ across traits.

7.3 Policy implications

The standard interpretation of $\text{PGI} \times \text{SES}$ gradients for BMI—that disadvantage “activates” genetic risk, and therefore interventions should target high-genetic-risk children in disadvantaged settings—rests on the H1 reading. The contribution of this paper to that debate is to show the H1 reading has been assumed rather than demonstrated: the gradient is equally consistent with running through family channels, and the screening rationale cannot stand on a gradient that has not been shown to be a direct-effect phenomenon.

A genetic-screening rationale requires the H1 reading to hold—that the children whose own alleles are most sensitive to disadvantage are identifiable and are the right targets. This paper shows that the widely cited $\text{PGI} \times \text{SES}$ gradient does not, on its own, supply that rationale: the gradient is equally consistent with operating through parental genotypes and household environments, in which case screening children by genetic risk would misidentify the operative channel. The gradient therefore should not be used to justify genetic stratification of children. It does not, by the same token, establish that household-targeted interventions are the answer; it shows that the question of which lever to pull is open. Universal interventions that improve household food environments (e.g., subsidised healthy meals, school nutrition programmes, parental nutrition education) are defensible on independent public-health grounds and do not depend on resolving the H1/H2 question;

allocating interventions by children’s polygenic risk does depend on it, and that question is not yet answered.

For mental health, the directional evidence—though not statistically confirmed—points in the opposite direction. The suggestive H1 pattern for SDQ raises the possibility that children’s own genetic endowment for wellbeing interacts with household resources. If confirmed in larger samples, interventions that directly support children’s cognitive and emotional development under disadvantage—such as school-based mental health programmes or early-childhood enrichment—would target the correct channel.

7.4 Limitations

What the family-GWAS comparison can deliver is itself bounded. The DGE PGI is a noisy proxy for the latent direct-effect score, and the near-zero DGE PGI $\times$ SES interaction does not rule out a direct-channel contribution that the current precision of the family-GWAS weights cannot detect. Imputed paternal genotypes inflate that noise further, and classical measurement error in either PGI attenuates interaction coefficients toward zero. The test of pure H1 in §6.3 cannot rule out the precision-penalty explanation of the Standard-vs-DGE divergence at Q5: the restriction $\beta_{GS}^{\text{Std}} = r \beta_{GS}^{\text{DGE}}$ is not rejected, so a pure-H1 process remains consistent with the data and the contrast does not decompose into channels. The analysis also does not apply ORIV-style measurement-error correction (Becker et al., 2021; Muslimova et al., 2024; Pereira and van Kippersluis, 2025), which requires a split-sample PGI construction unavailable here. As family-based GWAS samples grow (Tan et al., 2026; Alemu et al., 2025), the test of pure H1 will gain power to detect smaller direct-effect moderation.

Family GWAS also substitutes assumptions rather than weakening them. Conditioning on parental genotypes mechanically absorbs the parental indirect channel and most of the stratification, but the design adds three assumptions that population GWAS does not need: no sibling-mediated indirect effects, no extra-parental household channels, and no cross-locus assortative-mating linkage disequilibrium beyond what parental genotypes at SNP ℓ encode (Table 1). For BMI specifically, the existing literature bounds the bite of these substitutions: Howe et al. (2022) report a within-sibship Mendelian-randomisation slope of -5% (95% CI -14% to 4%), indistinguishable from zero shrinkage; Yengo et al. (2018a) find no detectable BMI signal of genome-wide assortative mating at the sample sizes then available. The three substituted assumptions are not testable within the MCS data, and on the strength of the cited evidence I treat each as second-order.

The constant-coefficient framework also commits to a population-mean reading of the direct effect. A strong biological reading of H1, under which the per-allele effect on metabolism or appetite genuinely varies with SES, would make $\beta_{\ell}^{\text{dir}}$ a function of SES. The family-GWAS weights estimate a mean over the joint distribution of SES and other moderators in the discovery sample, not the SES-specific parameter the strong-H1 reading targets. The DGE PGI $\times$ SES interaction in the MCS therefore identifies an SES gradient in this population-mean direct effect, which coincides with the SES-specific gradient when the per-allele effect is constant across environments and diverges from it when it is not. The MCS data cannot distinguish the two cases.

The Standard-vs-DGE comparison also localises the SES interaction to the family channel without isolating the specific household mechanism within it. The indirect-effect component encompasses parental dietary provision, health behaviours, economic decisions, and neighbourhood selection;

decomposing these sub-channels would require richer behavioural data than the MCS genetic sample provides.

The identifying variation supports inference only at the five MCS income quintiles. Counterfactuals at SES levels outside that support—extreme poverty or top-percentile wealth—are extrapolation. The family-based design also absorbs gene–environment correlation that operates through parental genotypes but not through the child’s own behavioural responses, peer effects correlated with the child’s genotype, or any channel that correlates g_{cil} with SES_i without passing through (g_{pit}, g_{mit}) .

Beyond the cohort itself, generalisability is bounded by institutional context. The MCS observes a single UK birth cohort from ages 5 to 17, and the UK welfare state provides universal healthcare, free school meals for low-income families, and relatively generous child benefits—features that may buffer some household-level channels. Whether the Standard-vs-DGE contrast looks the same in contexts with weaker safety nets is an open empirical question. Existing evidence suggests that gene–environment interaction patterns vary with institutional context: [Isungset et al. \(2022\)](#) find null or small gene–SES interactions for educational performance in the Norwegian welfare state, while [Houmark et al. \(2025\)](#) document a directional shift across developmental stages in Danish schools.

The position the analysis supports is therefore: for childhood BMI in the MCS, the SES gradient is present in the standard PGI’s prediction and absent from the DGE PGI’s; whether that reflects the indirect family channel or a direct channel the DGE PGI is too imprecise to recover is not identified, conditional on the maintained assumptions catalogued above. Deciding between the two requires either a larger family-GWAS discovery sample or a design that targets the SES-specific direct effect rather than its population mean.

7.5 Summary and future directions

This paper asks whether the well-documented socioeconomic gradient in the genetic prediction of BMI reflects environmental amplification of children’s own genetic effects or family-level channels embedded in standard polygenic indices. It introduces the Standard-versus-DGE design to answer that question, and reaches two results: the gradient is carried by the standard index and not by the direct-effects index, and the question of which channel that reflects is answerable in principle but not with a family GWAS of the size now available. The analysis proceeds in six steps. Step 1: the Standard PGI $\times$ SES interaction is negative and grows as parental genetic pathways are accounted for. Step 2: replacing the Standard PGI with the DGE PGI collapses the interaction to a coefficient indistinguishable from zero in every specification. Step 3: both PGIs enter a single joint regression. Step 4: pure H1 implies the two interactions stand in a fixed ratio $r = \sqrt{R_{Std}^2/R_{DGE}^2} \approx 10.7$; a cluster-robust Wald test of that restriction, $\beta_{GS}^{Std} = r \beta_{GS}^{DGE}$, is not rejected at any income quintile, because the DGE PGI is too imprecise once its interaction is scaled by r , so the test cannot decide between the channels. The main-effect ratio agrees with r only as an algebraic identity, not as an independent check. Step 5: three falsification probes characterise the contrast. The smoking null rules out the worry that the BMI pattern is a mechanical artefact of regressing any PGI on any socially patterned outcome; the SDQ contrast is qualitatively different from BMI; and the trio robustness exercise leaves the contrast intact. Step 6 synthesises: for BMI the SES gradient is established to be present in the standard PGI and absent from the DGE PGI, and deciding whether that reflects H1 or H2 is shown to require a larger family GWAS than now exists; for SDQ the cross-sectional evidence is underpowered; for smoking neither predictor generates a gradient.

The finding unsettles an active literature. What has been interpreted as genetic sensitivity to socioeconomic disadvantage—and cited as motivation for genetically-targeted interventions—has not, for BMI, been shown to be that: the gradient is equally consistent with socioeconomic variation in the family environments that parental genotypes help create. The motivation for genetically-targeted intervention is therefore not supported by this gradient, though identifying the operative channel awaits a larger family GWAS.

Several directions for future work follow directly from the limitations of the present analysis and from gaps in the broader literature.

The DGE PGI used here is itself an imperfect proxy for direct genetic effects. As family-based GWAS sample sizes grow and estimation precision improves (Tan et al., 2026; Howe et al., 2022), repeating the Standard-vs-DGE comparison with more powerful DGE weights will strengthen the test and potentially detect small direct-channel components that the current analysis cannot rule out.

Correcting for measurement error in the PGI using instrumental variable methods such as ORIV (Muslimova et al., 2024) would address the concern that attenuation bias in the interaction term may understate both direct-effect and family-channel moderation. The correction requires split-sample PGI construction that was not available for the present study.

If a family channel is what the gradient reflects, that channel is a composite: parental health behaviours, dietary provision, household organisation, and other pathways that covary with parental genotypes. Decomposing this channel—for example, by exploiting asymmetries between maternal and paternal PGI contributions, or by incorporating direct measures of parental behaviours and household food environments—would clarify which specific family mechanisms drive the socioeconomic gradient.

Extending the Standard-vs-DGE comparison to additional traits and cohorts would reveal whether the trait-specific pattern documented here (a robust contrast for BMI, a different and underpowered pattern for mental health, a null for smoking) replicates across institutional contexts and developmental stages, or whether it is specific to the UK cohort and age range studied.

Mostafavi et al. (2020) show that PGI prediction accuracy varies across socioeconomic strata even within an ancestry group. Quantifying whether and how this differential accuracy interacts with the Standard-vs-DGE comparison is necessary to fully separate measurement artefacts from genuine biological and social mechanisms.

Appendix

A. Polygenic index construction

This appendix documents the variant- and sample-level processing steps underlying the two BMI polygenic indices used in the main analysis. The construction of imputed parental genotypes, which feeds into the DGE PGI and into the parental controls in the main regressions, is described separately in Appendix 7.5.

Genotype data and pedigree. Child, maternal, and paternal genotypes for the UK Millennium Cohort Study were obtained as per-chromosome genotype filesets in standard binary format. Variant identifiers were harmonised to a common dbSNP 153 reference, duplicate variants were removed, and per-chromosome files were concatenated into a single genome-wide fileset aligned to human genome build GRCh38. A pedigree linking each child to its biological parents was constructed from the MCS family-linkage file, with missing parents flagged explicitly so that they could be either imputed or excluded depending on the analysis.

Standard BMI PGI. SNP weights were derived from the Yengo et al. (2018b) BMI GWAS summary statistics, the largest publicly available meta-analysis at the time of construction, combining the GIANT consortium discovery sample with UK Biobank replication. Summary statistics were coordinated with the MCS genotype panel using LDpred v1.0.10 (Vilhjálmsdóttir et al., 2015), restricted to HapMap3 variants, with the 1000 Genomes European sample as the linkage-disequilibrium reference panel and a frequency-discrepancy tolerance of 0.15 between the GWAS and the MCS panel. Posterior-mean effect sizes were obtained from LDpred’s Gibbs sampler under an infinitesimal prior, with several sparser priors estimated as sensitivity checks but not used in the final index. The shrunk weights were then applied to the harmonised MCS genotypes to produce a child-level BMI polygenic index.

Direct genetic effects (DGE) BMI PGI. SNP weights were drawn from the Tan et al. (2026) family-based BMI GWAS, which reports direct-effect estimates that have been purged of indirect parental and assortative-mating channels by conditioning on parental genotypes (Section 3.2.3). Of the 658,535 variants in the summary statistics, 585,466 with non-missing direct-effect estimates were retained. Strand-ambiguous SNPs (A/T and C/G) were excluded because allele orientation cannot be resolved unambiguously against the MCS panel, and variant coordinates were lifted over from GRCh37 to GRCh38 to align with the genotype build. The polygenic index was computed using the family-based scoring procedure of snipar (Young et al., 2022), which produces child, maternal, and paternal indices on a common scale within each family.

Estimation sample. The analysis sample of 5,446 children (Table 3) comprises children with non-missing BMI, non-missing SES, non-missing control variables, and a computable DGE PGI—that is, children for whom both parents were either directly genotyped or successfully imputed under the procedure described in Appendix 7.5. A subset of 2,707 children have both parents genotyped directly (the “full-trio” subsample used in the robustness analysis of Table 11); the remaining children have at least one imputed parent. All child, maternal, and paternal BMI PGIs are standardised to mean zero and unit standard deviation in the estimation sample.

B. Mendelian imputation of missing parental genotypes

The family-based identification strategy requires parental genotypes for two distinct purposes: as conditioning variables in the main regressions (the parental controls in Section 3.2.3), and as inputs to the family-based GWAS weights underlying the DGE PGI (Tan et al., 2026; Young et al., 2022). In the MCS, maternal genotypes are available for the great majority of genotyped children, but a substantial share of fathers were not genotyped. Two responses are possible: restrict the sample to fully observed child–mother–father trios, or recover the missing paternal genotypes using the Mendelian-transmission imputation procedure of Young et al. (2022). The main analysis adopts the second strategy and verifies that the central findings are reinforced—not artefactually generated—when the trio-only sample is used (Table 11). This appendix describes the imputation procedure, its statistical properties, and its interaction with the identification strategy.

Why imputation rather than trio-only estimation. Restricting attention to fully observed trios discards roughly half of the otherwise eligible children and, more importantly, induces selection on paternal participation. Father-genotyped families differ systematically from father-missing families on measured socioeconomic characteristics (Table A2), and likely on unmeasured ones as well. A trio-only estimator therefore answers a different question—the gradient among children whose fathers were retained in the genotyped sample—and is vulnerable to selection bias on precisely the dimension

(socioeconomic status) that the paper studies. Imputation preserves the full eligible sample at the cost of additional sampling variance, while leaving the identification logic intact.

The imputed quantity. For each missing paternal genotype g_{pil} at SNP ℓ , the imputed value is the conditional expectation of the true paternal genotype given the available family information,

$$\hat{g}_{pil} = \mathbb{E}[g_{pil} \mid g_{cil}, g_{mil}, f_\ell], \quad (36)$$

where g_{cil} and g_{mil} denote the child’s and the mother’s allele counts and f_ℓ denotes the population minor-allele frequency. The conditional expectation in (36) is computed under the assumptions of Mendelian transmission—each parent independently transmits one of two alleles to the child with probability one half—and Hardy–Weinberg equilibrium for the untransmitted paternal allele. Given the child’s and mother’s genotypes, the maternally transmitted allele can in most configurations be identified, leaving the paternally transmitted allele as a known quantity and the untransmitted paternal allele as the only unobserved component; that residual allele is replaced by its population expectation f_ℓ . When both parents are missing for a child with one or more genotyped siblings, an analogous calculation conditions on the sibling configuration; children for whom no such information is available are excluded from the estimation sample.

Properties of the imputed PGI. Aggregating the imputation across SNPs by applying the same weights used to construct the standard or DGE PGI yields an imputed parental polygenic index that inherits the conditional-expectation structure of (36). By construction this index is unbiased for the true parental PGI in the sense that

$$\mathbb{E}[\hat{\text{PGI}}_{pi} \mid g_{ci}, g_{mi}] = \mathbb{E}[\text{PGI}_{pi} \mid g_{ci}, g_{mi}],$$

and it depends only on pre-determined genetic and pedigree information. In particular, the imputation does not use the BMI outcome or any socioeconomic covariate, so the imputed parental PGI is mean-independent of (y_i, S_i) given the family genetic information. This is the core reason that imputation does not threaten identification: the segregation residual that drives the within-family identification of δ_ℓ is constructed from the child’s own allele count and the parental conditioning set, and replacing the conditioning set with its conditional expectation under Mendelian inheritance preserves the conditional-mean restriction $\mathbb{E}[u_{i\ell} \mid \cdot] = 0$.

Operational consequence: precision, not bias. The cost of imputation is a loss of statistical precision rather than a shift in the estimand. Because $\hat{g}_{pil}$ averages over the unknown untransmitted paternal allele, it is a noisier proxy for g_{pil} than the directly observed allele count would be. This widens standard errors on paternal coefficients and, to a lesser extent, on coefficients that interact with paternal terms; it does not introduce a systematic shift in any coefficient. The trio-only robustness analysis confirms this prediction: estimates of the standard-PGI $\times$ SES interaction obtained on the fully observed trio sample are larger in magnitude than the full-sample estimates, consistent with imputation attenuating rather than manufacturing the moderation pattern (Table 11).

Composition of the analysis sample. Among the 5,446 children in the estimation sample, 2,707 have both parents genotyped directly and 2,739 have at least one parent recovered through imputation. Table A2 reports descriptive comparisons of these two subsamples on polygenic indices, socioeconomic status, and other covariates. The imputed-parent subsample is concentrated at the lower end of the socioeconomic distribution—roughly one in five imputed-parent children fall in the bottom income quintile, against fewer than one in ten in the trio sample—which is the reason for preferring the imputed-inclusive sample over a trio-only restriction. Excluding these families would deliver an estimand defined over a relatively narrow upper-SES slice of the population and would discard precisely the variation in socioeconomic status the paper studies.

C. Supplementary tables

Table A1: Child PGIs, socioeconomic status, and BMI (conventional OLS standard errors)

	Dependent variable: BMI_{it}						
	(1) PGI	(2) SES	(3) + Controls	(4) + PGI×SES	(5) + Parents	(6) + Parent×Child	(7) + Parent×SES
Panel A: Standard PGI							
Child PGI	0.921*** (0.025)		0.947*** (0.021)	0.956*** (0.021)	0.883*** (0.026)	0.889*** (0.026)	0.900*** (0.027)
SES _c		-0.303*** (0.020)	-0.189*** (0.016)	-0.189*** (0.016)	-0.181*** (0.016)	-0.180*** (0.016)	-0.176*** (0.016)
Mother PGI					0.148*** (0.025)	0.151*** (0.025)	0.145*** (0.025)
Father PGI					0.013 (0.028)	0.011 (0.028)	-0.027 (0.030)
Child PGI × SES _c				-0.042*** (0.014)	-0.040*** (0.014)	-0.028* (0.015)	-0.070*** (0.019)
Mother PGI × Child PGI						0.070*** (0.017)	0.070*** (0.017)
Father PGI × Child PGI						0.016 (0.020)	0.028 (0.020)
Mother PGI × SES _c							0.030 (0.018)
Father PGI × SES _c							0.096*** (0.022)
Observations	5,446	5,446	5,446	5,446	5,446	5,446	5,446
R^2	0.049	0.009	0.442	0.442	0.443	0.443	0.444
Panel B: DGE PGI							
Child DGE PGI	0.092*** (0.028)		0.142*** (0.035)	0.141*** (0.035)	0.096*** (0.034)	0.096*** (0.034)	0.095*** (0.034)
SES _c		-0.303*** (0.020)	-0.289*** (0.016)	-0.289*** (0.016)	-0.210*** (0.016)	-0.211*** (0.016)	-0.210*** (0.016)
Mother PGI					0.584*** (0.022)	0.584*** (0.022)	0.590*** (0.022)
Father PGI					0.428*** (0.026)	0.429*** (0.026)	0.406*** (0.028)
DGE PGI × SES _c				0.004 (0.016)	-0.005 (0.016)	-0.009 (0.016)	-0.008 (0.016)
Mother PGI × DGE PGI						-0.046** (0.022)	-0.048** (0.022)
Father PGI × DGE PGI						0.043 (0.026)	0.046* (0.026)
Mother PGI × SES _c							-0.028* (0.015)
Father PGI × SES _c							0.049** (0.020)
Observations	5,446	5,446	5,446	5,446	5,446	5,446	5,446
R^2	0.000	0.009	0.397	0.397	0.419	0.419	0.419

Notes: Appendix replication of Table 5 using conventional OLS standard errors in place of child-level clustered standard errors. Point estimates are identical to the clustered SE version; only standard errors and significance stars differ because OLS treats each child-wave observation as independent, understating uncertainty in this panel setting. The clustered SE version (Table 5) is the preferred specification. Significance: * 10%, ** 5%, *** 1%.

Table A2: Descriptive comparison of real-trio and imputed-parent subsamples within the estimation sample

	Real trio ($N = 2,707$)	Has imputed ($N = 2,739$)	Diff.	p-value
<i>Polygenic indices (child-level, mean)</i>				
Standard BMI PGI (child)	−0.085	+0.033	−0.119	< 0.001
DGE BMI PGI (child)	−0.008	+0.018	−0.027	0.33
Mother BMI PGI (standard)	−0.100	+0.000	−0.100	< 0.001
Father BMI PGI (standard)	−0.073	−0.024	−0.049	0.03
<i>Socioeconomic status</i>				
SES quintile (1–5, mean)	3.449	2.912	+0.537	< 0.001
Quintile 1 share (%)	8.8	20.7	−11.9	< 0.001
Quintile 5 share (%)	26.3	16.8	+9.5	< 0.001
<i>Demographics and BMI</i>				
Female (proportion)	0.507	0.513	−0.006	0.67
BMI mean across waves	19.27	19.65	−0.39	< 0.001
Waves per child	4.82	4.72	+0.10	< 0.001

Notes: Comparison of children in the full BMI estimation sample ($N = 5,446$) split into the real-trio subsample (both parents directly genotyped) and the imputed-parent subsample (Mendelian imputation of at least one parent required for inclusion). Polygenic indices are child-level z -scores standardised in the full genotyped sample before sample restrictions. SES quintile is the modified OECD equivalised household income quintile at baseline. BMI mean is averaged across the waves each child contributes. p -values from two-sample t -tests for continuous variables and χ^2 tests for proportions. The quintile 1 share and quintile 5 share rows report the percentage of each subsample falling in the lowest and highest income quintile respectively; the full five-quintile distribution differs between the two groups at $p < 0.001$ by a single χ^2 test.

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